
Gauge-Fixed Sinkhorn Algorithms for Semi-Relaxed Optimal Transport

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Abstract

Semi-relaxed optimal transport (SROT) differs from the classical optimal transport framework by enforcing only one marginal constraint, leaving the second marginal free. This formulation arises naturally in applications such as domain adaptation, multi-omics single-cell analysis, and graph dictionary learning, where one distribution serves as a fixed reference while the other is only partially characterized. While the unregularized problem admits a simple closed-form solution, it is non-differentiable and highly sensitive to outliers, which makes it unsuitable for gradient-based learning pipelines. This paper studies a doubly-regularized formulation that combines an entropic penalty on the joint coupling with a Kullback–Leibler divergence on the second marginal, the same problem considered in the discrete setting in Le et al. [2021], Fukunaga and Kasai [2022b] and recoverable as a specialization of the general scaling framework of Chizat et al. [2018]. The contribution is a canonical *gauge-fixed representative* of the corresponding semi-relaxed Sinkhorn iteration, obtained by exploiting the probability constraint on feasible couplings to fix a translation gauge in the log-domain dual. Strong duality, together with existence and uniqueness of the gauge-fixed dual potentials, is established on general Polish spaces, without finite-support assumptions. An explicit Hilbert-projective contraction is proved for the gauge-fixed operator and is transferred into uniform convergence of the log-domain potentials, geometric convergence of the dual objective value, and total-variation convergence of the primal coupling. A separate (α, β) -bias analysis quantifies the gap to the unregularized SROT value as a function of the regularization parameters, and yields an L^1 control toward the set of unregularized minimizers in the discrete setting. **ajouter resultats experimentaux.**

1 Introduction

Optimal transport (OT) has become a foundational tool for comparing probability distributions in machine learning, with applications ranging from generative modeling and domain adaptation to graph learning and single-cell genomics Peyré and Cuturi [2019], Courty et al. [2017]. Classical OT enforces an exact mass-conservation constraint on both marginals, which is too rigid in many practical settings. When one distribution serves as a fixed reference and the other is only partially observed, contaminated by outliers, or known up to a noisy estimate, requiring exact matching on both sides is unrealistic. Semi-relaxed optimal transport (SROT) addresses this asymmetry by enforcing the marginal constraint on a single side and penalizing deviations of the second marginal through a divergence, typically the Kullback–Leibler (KL) divergence Chizat et al. [2018]. This formulation arises naturally in domain adaptation under target shift, in graph dictionary learning where the prototype is fixed but the assignment is not Vincent-Cuaz et al. [2022], and in multi-omics single-cell analysis where one modality acts as an anchor for partial alignment.

37 Despite this conceptual appeal, the unregularized SROT problem is poorly suited to modern learning
38 pipelines. Because only one marginal is constrained, the infimum decouples across the source space
39 into independent pointwise minimizations, and an optimal coupling can be supported on a transport
40 map of the form $x \mapsto \arg \min_y c(x, y)$. This closed form has three serious consequences. First, the
41 resulting cost is non-differentiable almost everywhere with respect to the cost matrix or the source
42 measure, which prevents its use as a loss inside gradient-based architectures. Second, the pointwise
43 structure makes the solution acutely sensitive to outliers in the support of Q : a single anomalous
44 target point with low cost can absorb arbitrarily large mass. Third, the decoupling discards any global
45 matching geometry, since the assignment of mass at x ignores the assignments at all other source
46 locations Le et al. [2021], Nguyen et al. [2023].

47 To overcome these limitations, this paper studies a doubly-regularized SROT problem combining an
48 entropic penalty (strength $\alpha > 0$) on the joint coupling with a Kullback–Leibler divergence (strength
49 $\beta > 0$) on the second marginal. As shown below, both regularizers are strictly necessary: neither
50 alone yields a strictly convex, globally coupled, differentiable formulation amenable to gradient-based
51 optimization. The same doubly-regularized formulation has been studied in the discrete setting under
52 the names robust semi-constrained optimal transport Le et al. [2021] and SR-Sinkhorn Fukunaga and
53 Kasai [2022b], and is recoverable as a specialization of the general scaling framework of Chizat et al.
54 [2018]. The contribution of this paper is not a new alternating-scaling iteration but a canonical gauge-
55 fixed representative of the same projective orbit, together with the convergence and approximation
56 analysis it makes possible on general Polish spaces. The resulting dual problem, derived through
57 the variational formula of the KL divergence rather than through the general framework of Chizat
58 et al. [2018], is translation-invariant in the log-domain. This structure is exploited to define the
59 *Gauge-Fixed Sinkhorn* iteration, with a built-in normalization step and provable linear convergence.
60 A parallel normalization idea was identified for unbalanced OT in Séjourné et al. [2022]; the present
61 work develops its analogue for the semi-relaxed regime, where the asymmetry between the two
62 marginals gives rise to a different translation invariance.

63 **Contributions.** The contributions of this paper are summarized as follows.

- 64 • **A translation-invariant dual for doubly-regularized SROT.** A novel dual formulation is
65 derived, exploiting the fact that feasible primal couplings are probability measures, an aspect
66 not used in the more general derivation of Chizat et al. [2018]. The resulting log-domain
67 dual is translation-invariant and admits a natural gauge-fixing condition selecting a canonical
68 pair of potentials.
- 69 • **A canonical gauge-fixed representative of the SR/RS-Sinkhorn orbit.** The translation
70 invariance of the dual is used to select a canonical normalization of the standard semi-
71 relaxed Sinkhorn iteration of Le et al. [2021], Fukunaga and Kasai [2022b], which is also
72 recoverable as a specialization of the scaling framework of Chizat et al. [2018]. The resulting
73 Gauge-Fixed Sinkhorn updates are projectively equivalent to those iterations but carry the
74 gauge-fixing constant explicitly, yielding primal iterates that are automatically probability
75 measures at every step.
- 76 • **Hilbert-projective contraction transferred to L^∞ and total variation.** Strong duality
77 and existence and uniqueness of the gauge-fixed dual potentials are established on general
78 Polish spaces. A Hilbert-projective contraction is proved for the gauge-fixed operator, with
79 an explicit contraction factor stated in Theorem 5.2. Gauge fixing is what converts projective
80 convergence into uniform convergence of the log-domain potentials, which in turn transfers
81 to geometric convergence of the dual objective value and total-variation convergence of the
82 primal coupling.
- 83 • **Quantitative approximation error bounds.** Explicit upper bounds on the gap between the
84 regularized and unregularized SROT values are provided as functions of α and β separately,
85 with weak convergence of the regularized minimizers and an L^1 -type control in the discrete
86 setting under a mild row-non-degeneracy condition. The natural unregularized limit is the
87 restricted value $SROT_Q$ defined in equation (15), which coincides with $SROT$ under mild
88 support and continuity conditions on Q .

2 Related Work

Two lines of work are directly concerned with the same doubly-regularized semi-relaxed optimal transport problem studied here. Le et al. [2021] introduce robust semi-constrained optimal transport with entropic regularization and propose the Robust Semi-Sinkhorn algorithm in the finite-dimensional setting, together with computational-complexity and barycenter-computation results. Fukunaga and Kasai [2022b] study SR-Sinkhorn for SROT in the discrete probability-simplex setting, explicitly relating it to the Robust Semi-Sinkhorn iteration of Le et al. [2021], and derive bounds on the functional-value gap, the marginal-constraint gap, and the OT-distance gap. The alternating-scaling iteration considered in the present work is the same projective orbit; the contribution lies in selecting a canonical gauge-fixed representative of that orbit and analyzing it in Hilbert’s projective metric on general Polish spaces.

These same iterations also fit within the general scaling-algorithm framework of Chizat et al. [2018], which encompasses unbalanced, partial, and semi-relaxed OT under a unified generalized Sinkhorn scheme. The semi-relaxed problem studied here is recovered as a specialization, with the first marginal functional taken to be a hard constraint and the second a Kullback–Leibler divergence penalty toward a reference target measure. The convergence guarantees of Chizat et al. [2018] are stated in considerable generality, but they are not specialized to the probability-measure constraint inherent to the semi-relaxed setting and do not identify the gauge-fixing normalization exploited here.

The conceptual antecedent for the gauge-fixing idea is the translation-invariant Sinkhorn of Séjourné et al. [2022] for unbalanced OT, which identified that the slow convergence of Sinkhorn in the unbalanced regime stems from the absence of a global normalization of the iterates, equivalently a translation gauge in the dual potentials. The present work develops a parallel idea for the semi-relaxed problem, where the asymmetry between the two marginals (one hard-constrained, one penalized) gives rise to a different translation invariance, and where the gauge-fixed updates carry the normalization constant explicitly through the iteration.

Foundational results on entropic OT Cuturi [2013], Altschuler et al. [2017], Peyré and Cuturi [2019] and on the consistency of entropic regularization Carlier et al. [2017], Pal [2024] provide the broader analytical context, while general convergence properties of unbalanced Sinkhorn have been refined in Pham et al. [2020], Nguyen et al. [2023]. Other related lines include the unregularized SROT problem solved by block-coordinate Frank–Wolfe Fukunaga and Kasai [2022a], the KKT-based reduction of semi-unbalanced OT to balanced OT Liu et al. [2025], the semi-relaxed Gromov–Wasserstein divergence of Vincent-Cuaz et al. [2022] which provides one of the main motivating applications, partial OT and robust matching Nguyen et al. [2024], Truong et al. [2026], Yoon et al. [2026], Mignon et al. [2025], and alternative regularizations Blondel et al. [2018], Tang et al. [2024], Pele and Werman [2009], all of which are conceptually orthogonal to the gauge-fixing refinement pursued here.

3 Background and Problem Formulation

3.1 Preliminaries

Throughout this work, $(\mathcal{X}, P)$ and $(\mathcal{Y}, Q)$ are fixed probability spaces. The function $c : \mathcal{X} \times \mathcal{Y} \rightarrow \mathbb{R}$ is a lower semi-continuous measurable function, and α and β are strictly positive numbers called regularization parameters. If a result needs further assumptions to be proved, these will be given in the statement of the result. If not, previous assumptions will not be repeated. The notation $\Pi(P, \star)$ denotes the set of probability measures on $\mathcal{X} \times \mathcal{Y}$ where only the first marginal is constrained to be P .

3.2 Semi-Relaxed Optimal Transport

Consider the semi-relaxed optimal transport problem

$$\text{SROT} \triangleq \inf_{\pi \in \Pi(P, \star)} \int_{\mathcal{X} \times \mathcal{Y}} c(x, y) d\pi(x, y). \quad (1)$$

Because only one marginal is constrained, the infimum decouples across the source space into independent pointwise minimizations: an optimal coupling is supported on the transport map $x \mapsto \arg \min_{y \in \mathcal{Y}} c(x, y)$, so the optimal value reduces to $\int_{\mathcal{X}} \min_{y \in \mathcal{Y}} c(x, y) dP(x)$. Intuitively, since

the marginal constraint on $\mathcal{Y}$ is totally relaxed, all mass from a source point x trivially flows to the target y that minimizes the transport cost. While this closed form is analytically simple, it has three consequences that make the unregularized problem ill-suited to modern learning pipelines. First, the cost is non-differentiable almost everywhere with respect to the cost matrix or the source measure, preventing its use as a loss inside gradient-based architectures. Second, the pointwise structure makes the solution acutely sensitive to outliers in the support of Q , as a single anomalous target point with low cost can absorb arbitrarily large mass. Third, the decoupling discards any global matching geometry, since the assignment of mass at x ignores the assignments at all other source locations Le et al. [2021], Nguyen et al. [2023].

This paper therefore studies a doubly-regularized variant of (1) combining an entropic penalty on the joint coupling (strength $\alpha > 0$) with a KL divergence on the second marginal with respect to Q (strength $\beta > 0$). Both regularizers are strictly necessary. At $\beta = 0$, the second marginal remains free to concentrate arbitrarily and the objective separates into independent log-sum-exp minimizations on each fiber $\{x\} \times \mathcal{Y}$, losing the global geometry over the source space. Without entropic smoothing, a KL penalty in β alone yields neither tractable Sinkhorn-type updates nor a differentiable cost. Only the joint presence of both penalties produces a strictly convex, differentiable, globally coupled formulation. As shown in Section 4.1, the resulting dual, derived through the variational formula of the KL divergence rather than through the general framework of Chizat et al. [2018], is translation-invariant in the log-domain, a structure exploited to derive the *Gauge-Fixed Sinkhorn* algorithm with a built-in normalization step parallel to Séjourné et al. [2022].

3.3 Doubly-Regularized Semi-Relaxed Optimal Transport

Building on the motivation above, this paper studies the following doubly-regularized variant of (1):

$$\text{SROT}_{\alpha,\beta} \triangleq \inf_{\pi \in \Pi(P, \star)} P_{\alpha,\beta}(\pi) \triangleq \int_{\mathcal{X} \times \mathcal{Y}} c(x, y) d\pi(x, y) + \alpha D_{\text{KL}}(\pi \| P \otimes Q) + \beta D_{\text{KL}}(\pi_{\#}^{\mathcal{Y}} \| Q). \quad (2)$$

The entropic term $\alpha D_{\text{KL}}(\pi \| P \otimes Q)$ smooths the coupling and penalizes mass concentration, promoting soft and statistically robust assignments. The marginal term $\beta D_{\text{KL}}(\pi_{\#}^{\mathcal{Y}} \| Q)$ can be interpreted as a fidelity penalty that prevents the second marginal from drifting far from the reference measure Q , or equivalently, as a soft enforcement of the otherwise absent second marginal constraint. It can also be viewed as a regularization on the degree to which the model is allowed to reallocate mass across the target space.

This formulation fits within the general scaling framework of Chizat et al. [2018], recovered as a particular instance of [Chizat et al., 2018, Equation 3.1] with $n = 1$, where the functional $\mathcal{F}_1$ is the indicator of $\{P\}$ and $\mathcal{F}_2$ equals $\beta D_{\text{KL}}(\cdot \| Q)$, with the reference measures therein identified as P and Q respectively. The same doubly-regularized formulation, under different notation, appears in Le et al. [2021] as Robust Semi-constrained Optimal Transport and is studied in the discrete setting in Fukunaga and Kasai [2022b], where complexity bounds on the functional value, marginal, and OT distance gaps are established for the SR-Sinkhorn algorithm. The present work studies the same problem and the same alternating-scaling orbit as Le et al. [2021], Fukunaga and Kasai [2022b], and complements those works by exploiting the translation invariance of the dual problem to select a canonical gauge-fixed representative of that orbit, for which a Hilbert-projective contraction is proved on general Polish spaces, without finite-support assumptions.

4 Translation-Invariant Duality and Gauge-Fixed Sinkhorn

4.1 Translation-Invariant Dual Formulation

An alternative dual formulation is derived here with the goal of strengthening convergence guarantees to general Polish spaces, without finite-support assumptions. The derivation of Chizat et al. [2018] is highly general and thus does not exploit the fact that feasible primal couplings are necessarily probability measures in the SROT problem. The approach taken here relies on the variational formula

182 of the KL divergence and yields the following dual formulation

$$\sup_{u \in L^\infty(\mathcal{X}), v \in L^\infty(\mathcal{Y})} D_{\alpha, \beta}(u, v) \triangleq \int_{\mathcal{X}} u(x) dP(x) - \beta \log \left(\int_{\mathcal{Y}} e^{-\frac{v(y)}{\beta}} dQ(y) \right) - \alpha \log \left(\int_{\mathcal{X} \times \mathcal{Y}} e^{\frac{u(x) + v(y) - c(x, y)}{\alpha}} dP \otimes Q(x, y) \right). \quad (3)$$

183 A key structural property of $D_{\alpha, \beta}$ is that it is invariant under independent constant shifts of both
 184 potentials: for all $M, N \in \mathbb{R}$, $D_{\alpha, \beta}(u + M, v + N) = D_{\alpha, \beta}(u, v)$ (Lemma B.1). This is a direct
 185 consequence of the semi-relaxed structure: the linear term $\int_{\mathcal{X}} u(x) dP(x)$ contributes $+M$, the
 186 log-sum-exp term in v contributes $+N$, and the joint log-sum-exp contributes $-(M + N)$, yielding
 187 zero net change. This two-parameter gauge freedom is what the title of the subsection refers to, and it
 188 is the structural mechanism exploited to derive the Gauge-Fixed Sinkhorn updates below.

189 This translation invariance is conceptually related to, but structurally distinct from, the normalization
 190 insight of Séjourné et al. [2022]. In the unbalanced OT setting of Séjourné et al. [2022], both
 191 marginal terms in the dual are non-linear in the potentials (log-sum-exp rather than linear), so the
 192 dual functional is not natively translation-invariant; invariance must be restored by constructing an
 193 envelope functional optimized over a one-parameter translation. In the present semi-relaxed setting,
 194 the hard constraint on the first marginal yields a linear term $\int_{\mathcal{X}} u(x) dP(x)$ in the dual, and translation
 195 invariance holds directly for any independent shifts M and N , without any envelope construction.
 196 The spirit of the approach is the same: fixing the gauge of the dual potentials at each iteration yields
 197 a normalization step that is invisible in Hilbert’s projective metric yet produces iterates that are
 198 automatically probability measures and admit a clean contraction argument.

199 4.2 Sinkhorn-Like Alternating Projections

200 Two operators are introduced. They serve as the core updates of the alternating maximization scheme
 201 and, as established in the next section, they characterize the maximizers of the dual functional $D_{\alpha, \beta}$.

$$\Phi(v)(x) \triangleq -\alpha \log \left(\int_{\mathcal{Y}} e^{\frac{v(y) - c(x, y)}{\alpha}} dQ(y) \right), \quad (4)$$

$$\begin{aligned} \Psi(u)(y) &\triangleq \beta \log \left(\int_{\mathcal{X}} \left(\int_{\mathcal{Y}} e^{\frac{u(x) - c(x, y')}{\alpha}} dP(x) \right)^{\frac{\alpha}{\alpha + \beta}} dQ(y') \right) \\ &\quad - \frac{\alpha \beta}{\alpha + \beta} \log \left(\int_{\mathcal{X}} e^{\frac{u(x) - c(x, y)}{\alpha}} dP(x) \right). \end{aligned} \quad (5)$$

202 Following the classical Sinkhorn algorithm Cuturi [2013], the Gauge-Fixed Sinkhorn relies on
 203 alternating maximization of $D_{\alpha, \beta}$ over u and v in turn. The specific form of Φ and Ψ is dictated
 204 by the optimality conditions of the gauge-normalized dual: setting the partial derivatives of $D_{\alpha, \beta}$
 205 (Lemma B.2) to zero yields $u = \Phi(v) + \alpha \log Z(u, v)$ and a symmetric condition on v , where
 206 $Z(u, v) \triangleq \int_{\mathcal{X} \times \mathcal{Y}} e^{(u(x) + v(y) - c(x, y))/\alpha} dP \otimes Q(x, y)$. The translation invariance of $D_{\alpha, \beta}$ established
 207 in Lemma B.1 means that the gauge can be fixed to $Z(u, v) = 1$ without loss of generality, which
 208 immediately collapses the optimality conditions to $u = \Phi(v)$ and $v = \Psi(u)$. As a result, the operators
 209 automatically produce iterates that satisfy this normalization at every step, guaranteeing that the
 210 associated primal measure is a probability distribution with no additional renormalization required.
 211 The alternating maximization directly yields the unscaled iterative updates

$$u^{(l+1)} = \Phi(v^{(l)}); \quad (6a)$$

$$v^{(l+1)} = \Psi(u^{(l+1)}), \quad (6b)$$

212 where Φ is defined in (4) and Ψ in (5). For numerical stability, an equivalent scaled formulation is
 213 preferred in practice, which is presented in Algorithm 1 below. The scaled iterates are obtained from
 214 the unscaled updates (6) by setting $(a^{(l)}, b^{(l)}) \triangleq (e^{u^{(l)}/\alpha}, e^{v^{(l)}/\alpha})$.

215 The scaled iterations rely on the operators $\Xi : L^\infty(\mathcal{Y}) \rightarrow L^\infty(\mathcal{X})$ and $\Gamma : L^\infty(\mathcal{X}) \rightarrow L^\infty(\mathcal{Y})$,
 216 defined as

$$\Xi(b)(x) \triangleq \frac{1}{\int_{\mathcal{Y}} b(y) e^{-\frac{c(x,y)}{\alpha}} dQ(y)}; \quad (7)$$

$$\Gamma(a)(y) \triangleq \left(\frac{1}{\int_{\mathcal{X}} a(x) e^{-\frac{c(x,y)}{\alpha}} dP(x)} \right)^{\frac{\beta}{\alpha+\beta}} \times \left(\int_{\mathcal{Y}} \left(\int_{\mathcal{X}} a(x) e^{-\frac{c(x,y')}{\alpha}} dP(x) \right)^{\frac{\alpha}{\alpha+\beta}} dQ(y') \right)^{\frac{\beta}{\alpha}}, \quad (8)$$

217 which yield the scaled iterative updates

$$a^{(l+1)} = \Xi(b^{(l)}), \quad (9a)$$

$$b^{(l+1)} = \Gamma(a^{(l+1)}). \quad (9b)$$

Algorithm 1: Gauge-Fixed Sinkhorn (scaled)

Input: cost c ; reference measures P, Q ; regularization parameters $\alpha, \beta > 0$; stopping criterion (e.g., an error threshold or a maximum number of iterations).

Output: scaled potentials $a^{(l)}, b^{(l)}$.

Initialize $b^{(0)} \leftarrow \mathbf{1}$ and $l \leftarrow 0$;

while the stopping criterion is not met **do**

$a^{(l+1)} \leftarrow \Xi(b^{(l)});$
 $b^{(l+1)} \leftarrow \Gamma(a^{(l+1)});$
 $l \leftarrow l + 1;$

return $a^{(l)}, b^{(l)}$;

218 Compared to the standard SR/RS-Sinkhorn iterates of Le et al. [2021], Fukunaga and Kasai [2022b],
 219 also obtained as a specialization of the scaling framework of [Chizat et al., 2018, Equation 3.6], the
 220 operator Ψ contains an additional normalization term (the first summand in (5)). The two sets of
 221 iterates are projectively equivalent: they differ only by a constant translation in v , and the translation
 222 invariance of $D_{\alpha,\beta}$ ensures that both yield the same dual objective value at every step. The advantage
 223 of the gauge-fixed form is that it admits a clean contraction analysis in Hilbert’s projective metric, in
 224 which constant rescalings are invisible, and produces iterates that are automatically normalized. The
 225 practical benefits of this normalization are confirmed experimentally in Section 6.

226 5 Main Results

227 5.1 Existence and Uniqueness

228 The following theorem establishes attainment of the dual supremum together with the existence
 229 and uniqueness of a maximizing pair satisfying the gauge-fixed optimality conditions. The proof of
 230 attainment, along with the supporting technical lemmas, is given in Appendix B, while the existence
 231 and uniqueness of the gauge-fixed pair are established in Appendix C.

232 **Theorem 5.1.** *The supremum in problem (3) is attained. Moreover, there exists a unique pair*
 233 *$(\hat{u}, \hat{v}) \in L^\infty(\mathcal{X}) \times L^\infty(\mathcal{Y})$ satisfying*

$$\hat{u} = \Phi(\hat{v}), \quad (10)$$

$$\hat{v} = \Psi(\hat{u}), \quad (11)$$

234 *and this pair maximizes $D_{\alpha,\beta}$ in (3).*

235 The existence of a maximizer is established by restricting maximizing sequences to uniformly
 236 bounded potentials, which allows for the extraction of weak-* convergent subsequences via the

237 Banach–Alaoglu theorem. The global supremum is then attained due to the weak-* lower semi-
238 continuity of the integral functionals defining the dual objective $D_{\alpha,\beta}$.

239 The conditions (10) and (11) are precisely the gauge-normalized first-order optimality conditions of
240 $D_{\alpha,\beta}$: they arise from setting its partial derivatives (Lemma B.2) to zero and simultaneously enforcing
241 $Z(\hat{u}, \hat{v}) = 1$. The translation invariance of $D_{\alpha,\beta}$ is what makes this gauge-fixing canonical: every
242 orbit of translations $(u + M, v + N)$ contains exactly one pair satisfying (10) and (11), as stated in
243 Theorem 5.1.

244 Consequently, the infimum in the primal problem (2) is attained at a unique coupling $\hat{\pi} \ll P \otimes Q$,
245 whose Gibbs-type density is given by $d\hat{\pi}/d(P \otimes Q) = \exp((\hat{u}(x) + \hat{v}(y) - c(x, y))/\alpha)$. The formal
246 statement and proof are provided in Appendix C.

247 5.2 Linear Convergence of the Algorithm

248 In the classical unbalanced Sinkhorn setting with two relaxed marginals, [Chizat et al., 2018, The-
249 orem 3.11] establishes a linear rate by viewing each scaling update as a z -homogeneous order-
250 preserving map ($0 < z < 1$), which is strictly contractive in the Thompson metric. In the present
251 semi-relaxed setting this argument breaks down: the update Γ in (9b) includes a global normalization
252 factor that is spatially constant but iterate-dependent, destroying the required z -homogeneity. Con-
253 vergence guarantees of complexity type already exist for the same orbit in the discrete setting, with
254 bounds on the functional-value, marginal, and OT-distance gaps established for SR-Sinkhorn in Fuku-
255 naga and Kasai [2022b]. The contribution of this section is to derive a Hilbert-projective contraction
256 for the gauge-fixed representative on general Polish spaces, from which uniform convergence of the
257 gauge-fixed log-potentials, geometric convergence of the dual objective value, and total-variation
258 convergence of the primal coupling all follow as consequences. The Hilbert projective metric is
259 the natural tool for this, as it is invariant under multiplication by positive scalars [Lemmens and
260 Nussbaum, 2014, Lemma 2.1(iv)], making such global rescalings invisible at the level of distances.

261 Convergence is quantified in Hilbert’s projective metric d_H on the positive cones $L_{++}^\infty(\mathcal{X})$ and
262 $L_{++}^\infty(\mathcal{Y})$ of essentially bounded functions with positive essential infimum, in the sense of Birkhoff’s
263 version on cones [Lemmens and Nussbaum, 2014, Equation (2.2)].

264 **Theorem 5.2.** *Let $(\hat{u}, \hat{v})$ be the unique maximizing pair for (3) satisfying (10) and (11), and let*
265 *$\hat{a} \triangleq e^{\hat{u}/\alpha}$, $\hat{b} \triangleq e^{\hat{v}/\alpha}$. The scaled iterates of Algorithm 1 satisfy, for all $l \geq 0$,*

$$d_H(b^{(l)}, \hat{b}) \leq (\rho_{(c, \alpha, \beta)})^l d_H(b^{(0)}, \hat{b}), \quad (12)$$

$$d_H(a^{(l+1)}, \hat{a}) \leq \tanh\left(\frac{\|c\|_\infty}{\alpha}\right) (\rho_{(c, \alpha, \beta)})^l d_H(b^{(0)}, \hat{b}), \quad (13)$$

266 where d_H is Hilbert’s projective metric [Lemmens and Nussbaum, 2014, Equation (2.2)] and

$$\rho_{(c, \alpha, \beta)} \triangleq \frac{\beta}{\alpha + \beta} \tanh\left(\frac{\|c\|_\infty}{\alpha}\right)^2. \quad (14)$$

267 The proof is detailed in Appendix E. The proof rests on the contraction principle for positive linear
268 operators in Hilbert’s projective metric: the kernels associated with Ξ and Γ have finite projective
269 diameter and are therefore strict contractions for d_H by [Lemmens and Nussbaum, 2014, Theorem 2.9].
270 The normalization built into Γ identifies a complete slice of the positive cone on which the one-step
271 map $\Gamma \circ \Xi$ is a contraction with rate $\rho_{(c, \alpha, \beta)}$, and Banach’s fixed-point theorem then yields the
272 geometric convergence.

273 Note that initializing the scaled algorithm with $b^{(0)} = 1$ is a natural and canonical choice. Indeed,
274 since $\hat{b} = e^{\hat{v}/\alpha}$, the initial Hilbert distance simplifies to $d_H(1, \hat{b}) = \text{osc}(\hat{v})/\alpha$, explicitly quantifying
275 how a smaller regularization parameter α increases the initial distance to the fixed point.

276 This linear rate convergence transfers directly to the dual objective value (see Corollary E.2), the
277 unscaled log-domain potentials $(u^{(l)}, v^{(l)})$ in L^∞ (see Corollary E.1), and the associated primal
278 iterates in total variation (see Corollary E.3). All formal statements and their respective proofs are
279 detailed in Appendix E.

5.3 Approximation Error

In classical entropic OT, the gap between the regularized and unregularized values has been studied in Carlier et al. [2017], Pal [2024]. The present section establishes an analogue for doubly-regularized SROT, quantifying the effect of α and β separately. A subtlety specific to the semi-relaxed setting is that the KL penalty on the second marginal forces the optimal coupling to be absolutely continuous with respect to $P \otimes Q$, so the regularized problem can only converge to unregularized minimizers whose second marginal is absolutely continuous with respect to Q . This motivates introducing the restricted value

$$SROT_Q \triangleq \inf \left\{ \int_{\mathcal{X} \times \mathcal{Y}} c(x, y) d\pi(x, y) : \pi \in \Pi(P, \star), \pi \ll P \otimes Q \right\}, \quad (15)$$

which satisfies $SROT_Q \geq SROT$ in general, with equality whenever the unregularized problem can be approximated by couplings absolutely continuous with respect to $P \otimes Q$. Note that $SROT_Q$ coincides with $SROT$ under mild conditions (e.g., full support of Q and continuous costs, or in the discrete setting with full support). This is formally stated and proven in Appendix D, see Lemma D.1. Intuitively, the gap $SROT_Q \geq SROT$ can only be strict when the pointwise minimizers in y lie in Q -null regions of $\mathcal{Y}$. Full support of Q rules this out, while continuity of $c(x, \cdot)$ ensures that near-minimizers form an open set, which therefore carries positive Q -mass and forces equality. The reference measure Q should therefore be chosen to have full support on $\mathcal{Y}$ whenever possible, as this ensures that the regularized values converge to the correct unregularized target. The importance of this choice is further underlined by the explicit bounds in the following theorem.

5.3.1 Convergence in Value

Theorem 5.3. *Recall the definitions of $SROT$ in (1), of $SROT_Q$ in (15), and of $SROT_{\alpha, \beta}$ in (2). The two following assertions hold.*

1. *Without any additional assumptions,*

$$SROT_{\alpha, \beta} \xrightarrow{\alpha, \beta \rightarrow 0} SROT_Q. \quad (16)$$

In particular, if $SROT_Q = SROT$ (for instance under the assumptions of Lemma D.1), then $SROT_{\alpha, \beta} \xrightarrow{\alpha, \beta \rightarrow 0} SROT$.

2. *Assume that there exists $\tilde{\pi} \in \Pi(P, \star)$ such that*

$$\int_{\mathcal{X} \times \mathcal{Y}} c(x, y) d\tilde{\pi}(x, y) = SROT, \quad D_{KL}(\tilde{\pi} \| P \otimes Q) < +\infty. \quad (17)$$

Then $D_{KL}(\tilde{\pi}_{\#}^{\mathcal{Y}} \| Q) < +\infty$ and

$$0 \leq SROT_{\alpha, \beta} - SROT \leq \alpha D_{KL}(\tilde{\pi} \| P \otimes Q) + \beta D_{KL}(\tilde{\pi}_{\#}^{\mathcal{Y}} \| Q). \quad (18)$$

The proof is detailed in Appendix D. The convergence (16) in the first assertion is established through two matching inequalities. The lower bound $SROT_{\alpha, \beta} \geq SROT_Q$ is immediate, since the entropic term $\alpha D_{KL}(\pi \| P \otimes Q)$ in (2) forces any coupling with finite regularized cost to be absolutely continuous with respect to $P \otimes Q$, which is precisely the feasible set defining $SROT_Q$ in (15). The matching upper bound $\limsup_{\alpha, \beta \rightarrow 0} SROT_{\alpha, \beta} \leq SROT_Q$ is obtained by evaluating the regularized objective $P_{\alpha, \beta}$ in (2) at a Gibbs kernel $\pi^{\tau} \propto e^{-c/\tau}$ and balancing the cost approximation against the KL penalties through the choice $\tau = \sqrt{\alpha + \beta}$. The quantitative bound (18) in the second assertion then follows directly by evaluating $P_{\alpha, \beta}$ at the candidate $\tilde{\pi}$ provided by hypothesis (17).

A discussion of the hypotheses and scope of each assertion follows. The convergence (16) requires no assumption beyond the standing ones and is the weakest of the two results: it establishes convergence to $SROT_Q$ defined in (15), not to the unregularized value $SROT$ defined in (1). The gap $SROT_Q - SROT$ closes to zero precisely under the conditions of Lemma D.1. The quantitative bound (18) strengthens this to an explicit $\mathcal{O}(\alpha + \beta)$ rate, but at the cost of requiring an unregularized $SROT$ minimizer $\tilde{\pi}$ that is absolutely continuous with respect to $P \otimes Q$; this bound separates the contributions of α and β through the respective KL divergences and vanishes as both parameters tend to zero. A

canonical setting in which the hypothesis (17) holds is the discrete case with finite $\mathcal{X}$ and $\mathcal{Y}$ and $Q(\{y_j\}) > 0$ for every j : every feasible coupling then has finite KL divergence against $P \otimes Q$, so the existence of an admissible $\tilde{\pi}$ in (17) is automatic.

Partial limits, in which only one of the two parameters is sent to zero, are deferred to Proposition D.1 in Appendix D. These regimes are of lesser practical relevance here, since the Sinkhorn algorithm developed in this paper requires both α and β to be strictly positive.

5.3.2 Convergence of the Minimizer

The value convergence established in Theorem 5.3 is a prerequisite for the weak convergence result below: the hypothesis $SROT_{\alpha_n, \beta_n} \rightarrow SROT$ holds whenever $SROT_Q = SROT$ and both parameters tend to zero, which is guaranteed in particular under the conditions of Lemma D.1. When $SROT_Q > SROT$ instead, that is, when Q lacks full support or the cost is discontinuous, the regularized values converge to $SROT_Q$ rather than $SROT$, and the hypothesis fails; in that case the same argument identifies weak limit points as minimizers of the restricted problem associated with $SROT_Q$.

Tightness of the family $(\hat{\pi}_{\alpha_n, \beta_n})_{n \geq 1}$ holds automatically when $\mathcal{Y}$ is compact, since every Borel probability measure on a compact metric space is tight [Billingsley, 1999, Theorem 5.1]. Uniform entropy bounds with respect to $P \otimes Q$ also imply tightness via a standard relative entropy argument [Billingsley [1999], Budhiraja and Dupuis [2019]].

Under the conditions above, every weak limit point of $(\hat{\pi}_{\alpha_n, \beta_n})_{n \geq 1}$ is a minimizer of the unregularized semi-relaxed problem (1), and the entire sequence converges weakly to π^* whenever this minimizer is unique. The formal statement and proof are deferred to Theorem D.1 in Appendix D.

This result is primarily of theoretical interest: it confirms that the regularized minimizers are consistent with the unregularized problem, but it provides neither a rate of convergence nor a norm-level control. In particular, the weak topology does not distinguish between minimizers when several exist. The quantitative complement below addresses both of these limitations in the discrete setting.

In finite spaces, a stronger L^1 control is available. For each row i , let $m_i \triangleq \min_j c(x_i, y_j)$ and $\Gamma_i \triangleq \{j : c(x_i, y_j) = m_i\}$. Assuming each row is non-constant, the row-separation gap

$$\Delta \triangleq \min_i \min_{j \notin \Gamma_i} (c(x_i, y_j) - m_i) \quad (19)$$

is strictly positive.

Theorem 5.4. Assume that $\mathcal{X} = \{x_1, \dots, x_n\}$ and $\mathcal{Y} = \{y_1, \dots, y_m\}$ are finite, with $p_i \triangleq P(\{x_i\})$ and $q_j \triangleq Q(\{y_j\}) > 0$ for all $j \in \{1, \dots, m\}$. Assume that each row of the cost matrix is non-constant. Define the set of unregularized minimizers

$$\mathcal{O} \triangleq \{\pi \in \Pi(P, \star) : \pi(x_i, y_j) = 0 \text{ for all } i \in \{1, \dots, n\}, j \notin \Gamma_i\}. \quad (20)$$

Then $\mathcal{O}$ is the set of minimizers of (1) and

$$\inf_{\pi \in \mathcal{O}} \|\hat{\pi}_{\alpha, \beta} - \pi\|_1 \leq \frac{2}{\Delta} (SROT_{\alpha, \beta} - SROT). \quad (21)$$

If each row admits a unique minimizer, then $\mathcal{O} = \{\pi^*\}$ and

$$\|\hat{\pi}_{\alpha, \beta} - \pi^*\|_1 \leq \frac{2}{\Delta} (SROT_{\alpha, \beta} - SROT) \leq \frac{2}{\Delta} \left(\alpha D_{KL}(\pi^* \| P \otimes Q) + \beta D_{KL}((\pi^*)_{\#}^{\mathcal{Y}} \| Q) \right). \quad (22)$$

The proof is detailed in Appendix D. The argument first uses the row-separation gap Δ in (19) to convert the primal cost excess $SROT_{\alpha, \beta} - SROT$ into a bound on the total mass that $\hat{\pi}_{\alpha, \beta}$ places outside the optimal columns $(\Gamma_i)_{i=1}^n$, and then projects $\hat{\pi}_{\alpha, \beta}$ onto $\mathcal{O}$ by reallocating each row's off-optimal mass to a single optimal column, which doubles the off-optimal mass into the L^1 distance and yields the factor $2/\Delta$; the second inequality then follows by applying the quantitative bound (18) of Theorem 5.3 with $\tilde{\pi} = \pi^*$.

Theorem 5.4 gives a key quantitative consequence of this analysis. In the discrete case, it simultaneously provides an explicit L^1 distance to the set of unregularized minimizers, an $\mathcal{O}(\alpha + \beta)$ rate

when a unique minimizer exists, and a separation of the contributions of α and β through the KL terms. The hypotheses (full support of Q , non-constant rows) are both necessary in a precise sense: dropping full support of Q allows $SROT_Q > SROT$, while a constant row prevents any L^1 control since the minimizer is then not isolated in the transport polytope. The row-separation gap Δ defined in (19) is a static constant determined solely by the ground-truth cost matrix c ; it depends neither on the regularization parameters α, β nor on the iterates produced by the algorithm. The convergence dynamics in Theorem 5.4 are therefore driven cleanly by $\alpha, \beta \rightarrow 0$ at the robust $\mathcal{O}(\alpha + \beta)$ rate, while $1/\Delta$ enters merely as a fixed geometric multiplier. This $1/\Delta$ dependency is not a flaw of the Gauge-Fixed Sinkhorn algorithm, but a fundamental property of exact, unregularized optimal transport, which is a linear program over the transport polytope. Geometrically, in exact OT, placing one unit of mass on a suboptimal target y_j with $j \notin \Gamma_i$ incurs a cost penalty of at least Δ relative to the optimal assignment, and the L^1 distance counts each unit of misplaced mass twice: once subtracted from the optimal target $y_{j_i} \in \Gamma_i$, once added to the suboptimal one. Consequently,

$$L^1 \text{ distance} \leq 2 \times (\text{misplaced mass}) \leq \frac{2}{\Delta} \times (\text{cost excess}). \quad (23)$$

This factor $2/\Delta$ is intrinsically sharp, as demonstrated by a toy example given in Appendix D at the end of the proof of Theorem 5.4.

Although the L^1 norm furnishes a universal discrete bound, it acts as a discrete metric and is therefore blind to the underlying geometry of the cost c , treating all suboptimal assignments on equal footing. Deriving a geometry-aware Wasserstein bound requires further structural assumptions on the cost; this limitation is discussed as a future direction in Section 7.

Combining the algorithmic and approximation results of this section yields a clean separation of the two error sources for the dual functional value. Geometric convergence of the dual objective value is established in Corollary E.2 from the Hilbert-projective contraction of Theorem 5.2, while the regularization bias is controlled by Theorem 5.3. Under the hypothesis (17) of an unregularized $SROT$ minimizer $\tilde{\pi}$ with finite KL divergence against $P \otimes Q$, the dual value evaluated along the iterates of Algorithm 1 therefore admits a decomposition of the form

$$|D_{\alpha,\beta}(u^{(l)}, v^{(l)}) - SROT| \leq \underbrace{C_{\text{opt}} \rho_{(c,\alpha,\beta)}^l}_{\text{optimization error}} + \underbrace{\alpha D_{KL}(\tilde{\pi} \| P \otimes Q) + \beta D_{KL}(\tilde{\pi}_{\#} \| Q)}_{\text{regularization bias}}, \quad (24)$$

where the first term is the geometric optimization error inherited from Theorem 5.2 (with $\rho_{(c,\alpha,\beta)}$ and C_{opt} as therein), and the second is the bias of the doubly-regularized objective with respect to the unregularized value, which separates the contributions of α and β explicitly. In particular, choosing α, β small reduces the bias at the cost of degrading the contraction factor, while increasing the iteration index l reduces the optimization error at fixed bias.

6 Experiments

CHECKER QUIL NY A PAS MON NOM DANS LE CODE. il faut bien preciser que la baseline que j'appelle "chizat" est aussi celle que les autres auteurs (les japonais) ont appelé SR Sinkhorn et RS Sinkhorn (les robust trucs). Notre algorithme offre des performances équivalentes ou légèrement supérieures à l'état de l'art sur des géométries euclidiennes simples. Cependant, il surpasse largement la baseline dans les régimes difficiles (faible régularisation entropique α , grandes dimensions, et coûts non-euclidiens), démontrant une bien meilleure stabilité numérique de la jauge. Dire qu'on a utilisé un nb fixe d'itérations A voir si on compare au solveur des japonais avec le franck block wolfe coordinate method, vu que ce n'est pas la même régularisation.

7 Conclusions and Limitations

A primary limitation of the present quantitative analysis is the reliance on the L^1 distance in Theorem 5.4, which is fundamentally blind to the geometry of the cost space. An important direction for future work is the derivation of a geometry-aware Wasserstein-distance control. This appears feasible if the static row-separation gap Δ is replaced by a Łojasiewicz-type growth condition on the cost, $c(x, y) - c(x, T(x)) \geq \kappa d_Y(y, T(x))^p$. The main obstacle for this extension in infinite dimensions is the possible singularity of the unregularized minimizer with respect to Q , whose careful treatment is left for future work.

Several other limitations should be made explicit. First, the contraction factor of Theorem 5.2 depends on the cost scale relative to the entropic regularization parameter α , and degenerates as $\alpha \rightarrow 0$; the proven rate is therefore informative in the moderately regularized regime rather than near the unregularized limit. A sharper, shift-invariant formulation of the contraction factor in terms of the cost oscillation rather than its sup-norm may be considered in a separate dedicated pass. Second, the simultaneous limit $\alpha, \beta \rightarrow 0$ is, in general, a limit toward the restricted value $SROT_Q$ rather than toward $SROT$ itself, with equality $SROT_Q = SROT$ guaranteed only under the support and continuity conditions discussed in Section 5.3; absent such conditions, the proven approximation results identify weak limit points as minimizers of the restricted problem. Third, the quantitative L^1 minimizer bound of Theorem 5.4 is finite-dimensional and depends on the row-separation gap Δ , so it is most informative when the cost matrix has well-separated row minima and degrades as the rows approach degeneracy. Finally, the experimental claims (Section 6) are limited to the regimes tested and should not be extrapolated to substantially different geometries, dimensions, or scales.

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510 A Weak Duality

511 QUAND MEME DANS L'ANNEXE IL FAUDRA AJOUTER DES PHRASES QUI EXPLIQUENT
 512 CE QU ON FAIT. PAS TRES COMPLIQUE MAIS SURTOUT DANS LA PARTIE SUR LE DUAL
 513 ATTAINMENT OU PAS MAL DE LEMMES SONT INTRODITS DANS L'ANNEXE.

514 **Lemma A.1.** *Weak duality holds between the primal problem in (2) and the dual problem in (3).*
 515 *More precisely,*

$$\sup_{u \in L^\infty(\mathcal{X}), v \in L^\infty(\mathcal{Y})} D_{\alpha, \beta}(u, v) \leq \inf_{\pi \in \Pi(P, \star)} P_{\alpha, \beta}(\pi) < +\infty. \quad (25)$$

516 *Proof:* Let $\pi \in \Pi(P, \star)$ be any feasible measure for problem (2); that is $P_{\alpha, \beta}(\pi) < +\infty$. Then $\pi \ll$
 517 $P \otimes Q$ and $\pi_{\#}^{\mathcal{Y}} \ll Q$. Let $(u, v) \in L^\infty(\mathcal{X}) \times L^\infty(\mathcal{Y})$ be arbitrary potentials. The Gibbs variational
 518 inequality for the Kullback–Leibler divergence, also known as the Donsker–Varadhan variational
 519 formula, see [Donsker and Varadhan, 1975, Lemma 2.1], yields, for all bounded measurable function
 520 h ,

$$D_{\text{KL}}(\mu || \nu) \geq \int h \, d\mu - \log \left(\int e^h \, d\nu \right), \quad (26)$$

521 where μ and ν are probability measures. Applying this inequality with

$$h(x, y) = \frac{u(x) + v(y) - c(x, y)}{\alpha},$$

522 $\mu = \pi$, and $\nu = P \otimes Q$, gives

$$\begin{aligned} \alpha D_{\text{KL}}(\pi || P \otimes Q) &\geq \int_{\mathcal{X} \times \mathcal{Y}} (u(x) + v(y) - c(x, y)) \, d\pi(x, y) \\ &\quad - \alpha \log \left(\int_{\mathcal{X} \times \mathcal{Y}} e^{\frac{u(x) + v(y) - c(x, y)}{\alpha}} \, dP \otimes Q(x, y) \right). \end{aligned} \quad (27)$$

523 Similarly, when $h(y) = -v(y)/\beta$, $\mu = \pi_{\#}^{\mathcal{Y}}$, and $\nu = Q$, the inequality in (26) yields

$$\beta D_{\text{KL}}(\pi_{\#}^{\mathcal{Y}} || Q) \geq - \int_{\mathcal{Y}} v(y) \, d\pi_{\#}^{\mathcal{Y}}(y) - \beta \log \left(\int_{\mathcal{Y}} e^{-\frac{v(y)}{\beta}} \, dQ(y) \right). \quad (28)$$

524 Adding the inequalities in (27) and (28) to the primal cost term $\int c(x, y) \, d\pi(x, y)$ yields

$$\begin{aligned} P_{\alpha, \beta}(\pi) &\geq \int_{\mathcal{X} \times \mathcal{Y}} (u(x) + v(y)) \, d\pi(x, y) - \int_{\mathcal{Y}} v(y) \, d\pi_{\#}^{\mathcal{Y}}(y) \\ &\quad - \alpha \log \left(\int_{\mathcal{X} \times \mathcal{Y}} e^{\frac{u(x) + v(y) - c(x, y)}{\alpha}} \, dP \otimes Q(x, y) \right) - \beta \log \left(\int_{\mathcal{Y}} e^{-\frac{v(y)}{\beta}} \, dQ(y) \right). \end{aligned} \quad (29)$$

525 Since $\pi_{\#}^{\mathcal{X}} = P$, it holds that

$$\int_{\mathcal{X} \times \mathcal{Y}} (u(x) + v(y)) d\pi(x, y) = \int_{\mathcal{X}} u(x) dP(x) + \int_{\mathcal{Y}} v(y) d\pi_{\#}^{\mathcal{Y}}(y). \quad (30)$$

526 Substituting (30) into (29) implies the cancellation of the terms involving v , and therefore

$$\begin{aligned} P_{\alpha, \beta}(\pi) &\geq \int_{\mathcal{X}} u(x) dP(x) - \alpha \log \left(\int_{\mathcal{X} \times \mathcal{Y}} e^{\frac{u(x) + v(y) - c(x, y)}{\alpha}} dP \otimes Q(x, y) \right) \\ &\quad - \beta \log \left(\int_{\mathcal{Y}} e^{-\frac{v(y)}{\beta}} dQ(y) \right) \end{aligned} \quad (31)$$

$$= D_{\alpha, \beta}(u, v). \quad (32)$$

527 Since the inequality in (31) holds for all feasible π and all bounded potentials (u, v) , it then follows
528 that

$$\sup_{u, v} D_{\alpha, \beta}(u, v) \leq \inf_{\pi \in \Pi(P, \star)} P_{\alpha, \beta}(\pi). \quad (33)$$

529 It remains to prove that the primal value, $P_{\alpha, \beta}(P \otimes Q)$, is finite. Since $P \otimes Q \in \Pi(P, \star)$ and
530 $D_{\text{KL}}(P \otimes Q || P \otimes Q) = 0$, it holds that

$$P_{\alpha, \beta}(P \otimes Q) = \int_{\mathcal{X} \times \mathcal{Y}} c(x, y) dP \otimes Q(x, y) < +\infty, \quad (34)$$

531 which follows from the boundedness of c . This completes the proof. ■

532 B Dual Attainment: Technical Lemmas

533 B.1 Preliminaries for the Proof of Dual Attainment

534 **Lemma B.1.** *For all functions $u \in L^{\infty}(\mathcal{X})$ and $v \in L^{\infty}(\mathcal{Y})$, and for all real numbers $M, N \in \mathbb{R}$,
535 the dual functional $D_{\alpha, \beta}$ in (3) satisfies*

$$D_{\alpha, \beta}(u + M, v + N) = D_{\alpha, \beta}(u, v). \quad (35)$$

536 *Proof:* Consider the functions $u \in L^{\infty}(\mathcal{X})$ and $v \in L^{\infty}(\mathcal{Y})$; and the reals $M, N \in \mathbb{R}$. Consider also
537 the problem in (3),

$$\begin{aligned} \sup_{u \in L^{\infty}(\mathcal{X}), v \in L^{\infty}(\mathcal{Y})} D_{\alpha, \beta}(u, v) &\triangleq \int_{\mathcal{X}} u(x) dP(x) - \beta \log \left(\int_{\mathcal{Y}} e^{-\frac{v(y)}{\beta}} dQ(y) \right) \\ &\quad - \alpha \log \left(\int_{\mathcal{X} \times \mathcal{Y}} e^{\frac{u(x) + v(y) - c(x, y)}{\alpha}} dP \otimes Q(x, y) \right) \end{aligned} \quad (36)$$

538 from which it follows that

$$\begin{aligned} D_{\alpha,\beta}(u+M, v+N) &= \int_{\mathcal{X}} u(x) dP(x) + M - \beta \log \left(\int_{\mathcal{Y}} e^{-\frac{v(y)+N}{\beta}} dQ(y) \right) \\ &\quad - \alpha \log \left(\int_{\mathcal{X} \times \mathcal{Y}} e^{\frac{u(x)+M+v(y)+N-c(x,y)}{\alpha}} dP \otimes Q(x,y) \right) \end{aligned} \quad (37)$$

$$\begin{aligned} &= \int_{\mathcal{X}} u(x) dP(x) + M - \beta \log \left(e^{-\frac{N}{\beta}} \int_{\mathcal{Y}} e^{-\frac{v(y)}{\beta}} dQ(y) \right) \\ &\quad - \alpha \log \left(e^{\frac{M+N}{\alpha}} \int_{\mathcal{X} \times \mathcal{Y}} e^{\frac{u(x)+v(y)-c(x,y)}{\alpha}} dP \otimes Q(x,y) \right) \end{aligned} \quad (38)$$

$$\begin{aligned} &= \int_{\mathcal{X}} u(x) dP(x) + M - (M+N) + N - \beta \log \left(\int_{\mathcal{Y}} e^{-\frac{v(y)}{\beta}} dQ(y) \right) \\ &\quad - \alpha \log \left(\int_{\mathcal{X} \times \mathcal{Y}} e^{\frac{u(x)+v(y)-c(x,y)}{\alpha}} dP \otimes Q(x,y) \right) \end{aligned} \quad (39)$$

$$= D_{\alpha,\beta}(u, v). \quad (40)$$

539 This completes the proof. $\blacksquare$

540 **Lemma B.2.** The dual functional $D_{\alpha,\beta}$ in (3) is Gâteaux differentiable on $L^\infty(\mathcal{X}) \times L^\infty(\mathcal{Y})$. More-
541 over, for all $(u, v) \in L^\infty(\mathcal{X}) \times L^\infty(\mathcal{Y})$ and all directions $(a, b) \in L^\infty(\mathcal{X}) \times L^\infty(\mathcal{Y})$, it holds
542 that

$$\left\langle \frac{\partial D_{\alpha,\beta}(u, v)}{\partial u}, a \right\rangle = \int_{\mathcal{X}} a(x) \left(1 - \frac{1}{Z(u, v)} \int_{\mathcal{Y}} e^{\frac{u(x)+v(y)-c(x,y)}{\alpha}} dQ(y) \right) dP(x); \quad (41)$$

$$\left\langle \frac{\partial D_{\alpha,\beta}(u, v)}{\partial v}, b \right\rangle = \int_{\mathcal{Y}} b(y) \left(\frac{e^{-\frac{v(y)}{\beta}}}{R(v)} - \frac{1}{Z(u, v)} \int_{\mathcal{X}} e^{\frac{u(x)+v(y)-c(x,y)}{\alpha}} dP(x) \right) dQ(y), \quad (42)$$

543 where

$$Z(u, v) \triangleq \int_{\mathcal{X} \times \mathcal{Y}} e^{\frac{u(x)+v(y)-c(x,y)}{\alpha}} dP \otimes Q(x, y), \quad (43)$$

$$R(v) \triangleq \int_{\mathcal{Y}} e^{-\frac{v(y)}{\beta}} dQ(y). \quad (44)$$

544 *Proof:* Consider $(u, v) \in L^\infty(\mathcal{X}) \times L^\infty(\mathcal{Y})$ and $(a, b) \in L^\infty(\mathcal{X}) \times L^\infty(\mathcal{Y})$. Since the functions
545 u, v, a, b and c are bounded, the quantities $Z(u, v)$ and $R(v)$ are finite and strictly positive.

546 First, consider the derivative of $D_{\alpha,\beta}(u, v)$ in (3) with respect to u . The linear term satisfies

$$\left. \frac{d}{dt} \right|_{t=0} \int_{\mathcal{X}} (u(x) + ta(x)) dP(x) = \int_{\mathcal{X}} a(x) dP(x). \quad (45)$$

547 The term involving $R(v)$ does not depend on u . Hence it remains to differentiate the term involving
548 $Z(u, v)$. By dominated convergence [Rudin, 1987, Theorem 1.34],

$$\left. \frac{d}{dt} \right|_{t=0} Z(u + ta, v) = \frac{1}{\alpha} \int_{\mathcal{X} \times \mathcal{Y}} a(x) e^{\frac{u(x)+v(y)-c(x,y)}{\alpha}} dP \otimes Q(x, y). \quad (46)$$

549 Therefore, the chain rule for the logarithm yields,

$$\left. \frac{d}{dt} \right|_{t=0} (-\alpha \log Z(u + ta, v)) = -\frac{1}{Z(u, v)} \int_{\mathcal{X} \times \mathcal{Y}} a(x) e^{\frac{u(x)+v(y)-c(x,y)}{\alpha}} dP \otimes Q(x, y) \quad (47)$$

$$= - \int_{\mathcal{X}} a(x) \left(\frac{1}{Z(u, v)} \int_{\mathcal{Y}} e^{\frac{u(x)+v(y)-c(x,y)}{\alpha}} dQ(y) \right) dP(x), \quad (48)$$

where the equality (48) in follows from the Fubini's theorem [Rudin, 1987, Theorem 8.8]. Combining (45) and (48) concludes the proof of (41).

It remains to compute the derivative of $D_{\alpha,\beta}(u, v)$ in (3) with respect to v . The term $\int_{\mathcal{X}} u \, dP$ does not depend on v . From the dominated convergence and the chain rule of the logarithm, it follows that

$$\left. \frac{d}{dt} \right|_{t=0} \left(-\beta \log \int_{\mathcal{Y}} e^{-\frac{v(y)+tb(y)}{\beta}} dQ(y) \right) = \frac{1}{R(v)} \int_{\mathcal{Y}} b(y) e^{-\frac{v(y)}{\beta}} dQ(y). \quad (49)$$

Similarly,

$$\left. \frac{d}{dt} \right|_{t=0} (-\alpha \log Z(u, v + tb)) = -\frac{1}{Z(u, v)} \int_{\mathcal{X} \times \mathcal{Y}} b(y) e^{\frac{u(x)+v(y)-c(x,y)}{\alpha}} dP \otimes Q(x, y) \quad (50)$$

$$= -\int_{\mathcal{Y}} b(y) \left(\frac{1}{Z(u, v)} \int_{\mathcal{X}} e^{\frac{u(x)+v(y)-c(x,y)}{\alpha}} dP(x) \right) dQ(y), \quad (51)$$

where the equality in (51) follows from Fubini's theorem [Rudin, 1987, Theorem 8.8]. Combining (49) and (51) concludes the proof of (42). This completes the proof. ■

Corollary B.1. *Let $(\hat{u}, \hat{v}) \in L^\infty(\mathcal{X}) \times L^\infty(\mathcal{Y})$, and let Φ and Ψ be the operators defined in (4) and (5). If $\hat{u}$ satisfies the optimality condition (10) (understood P -almost everywhere), then $\hat{u}$ maximizes $u \mapsto D_{\alpha,\beta}(u, \hat{v})$, where $D_{\alpha,\beta}$ is defined in (3). Similarly, if $\hat{v}$ satisfies (11) (understood Q -almost everywhere), then $\hat{v}$ maximizes $v \mapsto D_{\alpha,\beta}(\hat{u}, v)$. Moreover, if (10) and (11) are simultaneously satisfied, then $(\hat{u}, \hat{v})$ maximizes $(u, v) \mapsto D_{\alpha,\beta}(u, v)$.*

Proof: The functional $D_{\alpha,\beta}$ in (3) is concave and differentiable on $L^\infty(\mathcal{X}) \times L^\infty(\mathcal{Y})$ and in particular concave with respect to each variable when the other one is fixed. As a consequence, a sufficient condition for optimality is the vanishing of the first-order derivative [Zalinescu, 2002, Chapter 2.4].

Assume that $\hat{v} \in L^\infty(\mathcal{Y})$ is fixed and that $\hat{u}$ satisfies (10). By Lemma B.2, the first-order derivative of the map $u \mapsto D_{\alpha,\beta}(u, \hat{v})$ is given by (41). Injecting (10) into this expression shows that the derivative vanishes identically. By concavity, $\hat{u}$ therefore maximizes $u \mapsto D_{\alpha,\beta}(u, \hat{v})$.

Similarly, assume that $\hat{u} \in L^\infty(\mathcal{X})$ is fixed and that $\hat{v}$ satisfies (11). Using again Lemma B.2, the derivative of $v \mapsto D_{\alpha,\beta}(\hat{u}, v)$ is given by (42). Injecting (11) into this expression yields the cancellation of the two terms and thus the vanishing of the derivative. By concavity, $\hat{v}$ maximizes $v \mapsto D_{\alpha,\beta}(\hat{u}, v)$.

Finally, if (10) and (11) are satisfied simultaneously, then both partial derivatives vanish. Since $(u, v) \mapsto D_{\alpha,\beta}(u, v)$ is jointly concave, $(\hat{u}, \hat{v})$ is a global maximizer of $D_{\alpha,\beta}$. ■

The following lemma gives uniform oscillation bounds for the update operators.

Lemma B.3. *For every pair of functions (u, v) in $L^\infty(\mathcal{X}) \times L^\infty(\mathcal{Y})$, the operators Φ in (4) and Ψ in (5) satisfy*

$$\text{osc}(\Phi(v)) \leq 2\|c\|_\infty; \quad (52)$$

$$\text{osc}(\Psi(u)) \leq \frac{2\beta}{\alpha + \beta} \|c\|_\infty. \quad (53)$$

Proof: The proof is divided into two parts, corresponding to (52) and (53). Let v be a function in $L^\infty(\mathcal{Y})$. For all $(x, y) \in \mathcal{X} \times \mathcal{Y}$, it holds that

$$-\|c\|_\infty \leq -c(x, y) \leq \|c\|_\infty \quad (54)$$

$$\iff e^{-\frac{\|c\|_\infty}{\alpha}} e^{\frac{v(y)}{\alpha}} \leq e^{\frac{v(y)-c(x,y)}{\alpha}} \leq e^{\frac{\|c\|_\infty}{\alpha}} e^{\frac{v(y)}{\alpha}} \quad (55)$$

$$\iff e^{-\frac{\|c\|_\infty}{\alpha}} \int_{\mathcal{Y}} e^{\frac{v(y)}{\alpha}} dQ(y) \leq \int_{\mathcal{Y}} e^{\frac{v(y)-c(x,y)}{\alpha}} dQ(y) \leq e^{\frac{\|c\|_\infty}{\alpha}} \int_{\mathcal{Y}} e^{\frac{v(y)}{\alpha}} dQ(y) \quad (56)$$

$$\iff -\alpha \log \left(\int_{\mathcal{Y}} e^{\frac{v(y)}{\alpha}} dQ(y) \right) - \|c\|_\infty \leq \Phi(v)(x) \leq -\alpha \log \left(\int_{\mathcal{Y}} e^{\frac{v(y)}{\alpha}} dQ(y) \right) + \|c\|_\infty \quad (57)$$

where the inequalities in (56) follow by integrating with respect to Q in (56); and the inequalities in (57) follow by the logarithm and multiplying by $-\alpha$ in (56). From (57) it follows that $\text{osc}(\Phi(v)) \leq 2\|\mathbf{c}\|_\infty$, which concludes the proof of (52)

It remains to prove (53). Let u be a function in $L^\infty(\mathcal{X})$ and define

$$B_u(y) \triangleq \int_{\mathcal{X}} e^{\frac{u(x) - \mathbf{c}(x,y)}{\alpha}} dP(x), \quad (58)$$

$$C_u \triangleq \int_{\mathcal{Y}} B_u(y')^{\frac{\alpha}{\alpha+\beta}} dQ(y'). \quad (59)$$

From (5), it holds that

$$\Psi(u)(y) = \beta \log(C_u) - \frac{\alpha\beta}{\alpha+\beta} \log(B_u(y)). \quad (60)$$

Since the term $\beta \log(C_u)$ does not depend on y , it has zero oscillation. Hence,

$$\text{osc}(\Psi(u)) = \frac{\alpha\beta}{\alpha+\beta} \text{osc}(\log(B_u)). \quad (61)$$

For all $(x, y) \in \mathcal{X} \times \mathcal{Y}$, it holds that

$$-\|\mathbf{c}\|_\infty \leq -\mathbf{c}(x, y) \leq \|\mathbf{c}\|_\infty \quad (62)$$

$$\iff e^{-\frac{\|\mathbf{c}\|_\infty}{\alpha}} e^{\frac{u(x)}{\alpha}} \leq e^{\frac{u(x) - \mathbf{c}(x,y)}{\alpha}} \leq e^{\frac{\|\mathbf{c}\|_\infty}{\alpha}} e^{\frac{u(x)}{\alpha}} \quad (63)$$

$$\iff e^{-\frac{\|\mathbf{c}\|_\infty}{\alpha}} \int_{\mathcal{X}} e^{\frac{u(x)}{\alpha}} dP(x) \leq \int_{\mathcal{X}} e^{\frac{u(x) - \mathbf{c}(x,y)}{\alpha}} dP(x) \leq e^{\frac{\|\mathbf{c}\|_\infty}{\alpha}} \int_{\mathcal{X}} e^{\frac{u(x)}{\alpha}} dP(x) \quad (64)$$

$$\iff \log \left(\int_{\mathcal{X}} e^{\frac{u(x)}{\alpha}} dP(x) \right) - \frac{\|\mathbf{c}\|_\infty}{\alpha} \leq \log(B_u(y)) \leq \log \left(\int_{\mathcal{X}} e^{\frac{u(x)}{\alpha}} dP(x) \right) + \frac{\|\mathbf{c}\|_\infty}{\alpha}, \quad (65)$$

where the inequalities in (64) follow by integrating with respect to P ; and the inequalities in (65) follow by taking the logarithm. From (65), it follows that

$$\text{osc}(\log(B_u)) \leq \frac{2\|\mathbf{c}\|_\infty}{\alpha}. \quad (66)$$

Consequently,

$$\text{osc}(\Psi(u)) \leq \frac{\alpha\beta}{\alpha+\beta} \frac{2\|\mathbf{c}\|_\infty}{\alpha} = \frac{2\beta}{\alpha+\beta} \|\mathbf{c}\|_\infty, \quad (67)$$

which proves (53). This completes the proof. $\blacksquare$

The following lemma shows uniform bounds on the operators.

Lemma B.4. *For all pair of functions (u, v) in $L^\infty(\mathcal{X}) \times L^\infty(\mathcal{Y})$, the operators Φ in (4) and Ψ in (5) satisfy*

$$\|\Phi(v)\|_\infty \leq \|v\|_\infty + \|\mathbf{c}\|_\infty; \quad (68)$$

$$\|\Psi(u)\|_\infty \leq \frac{2\beta}{\alpha+\beta} \|\mathbf{c}\|_\infty. \quad (69)$$

593 *Proof:* The proof is divided into two parts, corresponding to (68) and (69). Let v be a function in
 594 $L^\infty(\mathcal{Y})$. For all $(x, y) \in \mathcal{X} \times \mathcal{Y}$, it holds that

$$-\frac{\|v\|_\infty + \|c\|_\infty}{\alpha} \leq \frac{v(y) - c(x, y)}{\alpha} \leq \frac{\|v\|_\infty + \|c\|_\infty}{\alpha} \quad (70)$$

$$\iff e^{-\frac{\|v\|_\infty + \|c\|_\infty}{\alpha}} \leq e^{\frac{v(y) - c(x, y)}{\alpha}} \leq e^{\frac{\|v\|_\infty + \|c\|_\infty}{\alpha}} \quad (71)$$

$$\iff e^{-\frac{\|v\|_\infty + \|c\|_\infty}{\alpha}} \leq \int_{\mathcal{Y}} e^{\frac{v(y) - c(x, y)}{\alpha}} dQ(y) \leq e^{\frac{\|v\|_\infty + \|c\|_\infty}{\alpha}} \quad (72)$$

$$\iff -\|v\|_\infty - \|c\|_\infty \leq \Phi(v)(x) \leq \|v\|_\infty + \|c\|_\infty, \quad (73)$$

595 where the inequalities in (72) follow by integrating with respect to Q , and the inequalities in (73)
 596 follow by taking the logarithm and multiplying by $-\alpha$. Hence,

$$\|\Phi(v)\|_\infty \leq \|v\|_\infty + \|c\|_\infty, \quad (74)$$

597 which proves (68).

598 It remains to prove (69). Let u be a function in $L^\infty(\mathcal{X})$ and note that the definition of Ψ in (5) implies

$$\int_{\mathcal{Y}} e^{-\Psi(u)(y)/\beta} dQ(y) = 1. \quad (75)$$

599 As the measure Q has total mass 1, the previous equality implies

$$\inf_{y \in \mathcal{Y}} e^{-\Psi(u)(y)/\beta} \leq 1, \text{ and } \sup_{y \in \mathcal{Y}} e^{-\Psi(u)(y)/\beta} \geq 1. \quad (76)$$

600 Monotonicity of the logarithm function yields

$$\sup \Psi(u) \geq 0 \quad \text{and} \quad \inf \Psi(u) \leq 0. \quad (77)$$

601 The interval $[\inf \Psi(u), \sup \Psi(u)]$ contains the origin. The distance from the origin to any point
 602 within a closed interval is bounded through the total interval length. The following inequalities result
 603 from this inclusion

$$\sup \Psi(u) \leq \text{osc}(\Psi(u)) \quad \text{and} \quad -\inf \Psi(u) \leq \text{osc}(\Psi(u)). \quad (78)$$

604 The definition $\|\Psi(u)\|_\infty = \max(\sup \Psi(u), -\inf \Psi(u))$ establishes the bound $\|\Psi(u)\|_\infty \leq$
 605 $\text{osc}(\Psi(u))$. Since the oscillation bound in (53) is established as $\frac{2\beta}{\alpha+\beta}\|c\|_\infty$, this concludes the
 606 proof of (69). ■

607 The following corollary reduces the maximization problem to uniformly bounded potentials.

608 **Corollary B.2.** *Let (u, v) be a pair of functions in $L^\infty(\mathcal{X}) \times L^\infty(\mathcal{Y})$. Then, there exists another*
 609 *pair (f, g) in $L^\infty(\mathcal{X}) \times L^\infty(\mathcal{Y})$ such that*

$$D_{\alpha, \beta}(u, v) \leq D_{\alpha, \beta}(f, g), \quad \|f\|_\infty, \|g\|_\infty \leq 2\|c\|_\infty, \quad (79)$$

610 where $D_{\alpha, \beta}$ is defined in (3).

611 *Proof:* Let (u, v) be a pair of functions in $L^\infty(\mathcal{X}) \times L^\infty(\mathcal{Y})$. From Corollary B.1, the function $\Phi(v)$
 612 maximizes the map $u \mapsto D_{\alpha, \beta}(u, v)$. Hence,

$$D_{\alpha,\beta}(u, v) \leq D_{\alpha,\beta}(\Phi(v), v). \quad (80)$$

Moreover, Lemma B.3 gives

$$\text{osc}(\Phi(v)) \leq 2\|c\|_\infty. \quad (81)$$

Therefore, there exists a constant $A \in \mathbb{R}$ such that the function $f \triangleq \Phi(v) + A$ satisfies

$$\|f\|_\infty \leq 2\|c\|_\infty. \quad (82)$$

By Lemma B.1, the dual functional is invariant under constant shifts of the potentials. Thus,

$$D_{\alpha,\beta}(\Phi(v), v) = D_{\alpha,\beta}(f, v). \quad (83)$$

Let now $g \triangleq \Psi(f)$. From Corollary B.1, the function g maximizes the map $v \mapsto D_{\alpha,\beta}(f, v)$. Hence,

$$D_{\alpha,\beta}(f, v) \leq D_{\alpha,\beta}(f, g). \quad (84)$$

Moreover, Lemma B.4 gives

$$\|g\|_\infty = \|\Psi(f)\|_\infty \leq \frac{2\beta}{\alpha + \beta}\|c\|_\infty \leq 2\|c\|_\infty. \quad (85)$$

Combining (80), (83), and (84) yields

$$D_{\alpha,\beta}(u, v) \leq D_{\alpha,\beta}(f, g). \quad (86)$$

Finally, (82) and (85) imply that

$$\|f\|_\infty, \|g\|_\infty \leq 2\|c\|_\infty. \quad (87)$$

This completes the proof. ■

B.2 Proof of Theorem 5.1

Let $(u_n) \subset L^\infty(\mathcal{X})$ and $(v_n) \subset L^\infty(\mathcal{Y})$ be maximizing sequences. It follows from Corollary B.2 that (u_n) and (v_n) can be chosen such that for all n , it holds that

$$\|u_n\|_\infty, \|v_n\|_\infty \leq 2\|c\|_\infty. \quad (88)$$

Since $\mathcal{X}$ and $\mathcal{Y}$ are Polish spaces, the spaces $L^1(P)$ and $L^1(Q)$ are separable. Hence, the closed balls of $L^\infty(\mathcal{X})$ and $L^\infty(\mathcal{Y})$ are weak* compact and metrizable. Application of the Banach–Alaoglu theorem [Brézis, 2011, Theorem 3.16] yields existence of subsequences $(u_{n_k})_k$ and $(v_{n_k})_k$ such that

$$u_{n_k} \rightharpoonup^* \bar{u} \text{ in } L^\infty(\mathcal{X}), \text{ and } v_{n_k} \rightharpoonup^* \bar{v} \text{ in } L^\infty(\mathcal{Y}). \quad (89)$$

In particular, the following weak* convergence also holds in $L^\infty(\mathcal{X} \times \mathcal{Y})$:

$$u_{n_k} + v_{n_k} - c \rightharpoonup^* \bar{u} + \bar{v} - c. \quad (90)$$

Moreover, for all $h \in L^1(\mathcal{X} \times \mathcal{Y}, P \otimes Q)$, Fubini's theorem implies that $x \mapsto \int_{\mathcal{Y}} h(x, y) dQ(y)$ belongs to $L^1(P)$ and $y \mapsto \int_{\mathcal{X}} h(x, y) dP(x)$ belongs to $L^1(Q)$.

Thus, it follows from the convexity of $\exp$ and the weak* lower semi-continuity of the integral functionals in the dual objective in (3) [Brézis, 2011, Theorem 3.7] that

$$\liminf_{k \rightarrow \infty} \int_{\mathcal{Y}} e^{-\frac{v_{n_k}(y)}{\beta}} dQ(y) \geq \int_{\mathcal{Y}} e^{-\frac{\bar{v}(y)}{\beta}} dQ(y); \text{ and} \quad (91)$$

$$\liminf_{k \rightarrow \infty} \int_{\mathcal{X} \times \mathcal{Y}} e^{\frac{u_{n_k}(x) + v_{n_k}(y) - c(x, y)}{\alpha}} dP \otimes Q(x, y) \geq \int_{\mathcal{X} \times \mathcal{Y}} e^{\frac{\bar{u}(x) + \bar{v}(y) - c(x, y)}{\alpha}} dP \otimes Q(x, y). \quad (92)$$

From the continuity and monotonicity of $\log$ on $(0, \infty)$ and the two previous inequalities, it follows that

$$\liminf_{k \rightarrow \infty} \log \left(\int_{\mathcal{Y}} e^{-\frac{v_{n_k}(y)}{\beta}} dQ(y) \right) \geq \log \left(\int_{\mathcal{Y}} e^{-\frac{\bar{v}(y)}{\beta}} dQ(y) \right); \text{ and} \quad (93)$$

$$\liminf_{k \rightarrow \infty} \log \left(\int_{\mathcal{X} \times \mathcal{Y}} e^{\frac{u_{n_k}(x) + v_{n_k}(y) - c(x, y)}{\alpha}} dP \otimes Q(x, y) \right) \geq \log \left(\int_{\mathcal{X} \times \mathcal{Y}} e^{\frac{\bar{u}(x) + \bar{v}(y) - c(x, y)}{\alpha}} dP \otimes Q(x, y) \right) \quad (94)$$

From which it can be deduced that

$$\begin{aligned} \sup_{u \in L^\infty(\mathcal{X}), v \in L^\infty(\mathcal{Y})} D_{\alpha, \beta}(u, v) &= \lim_{k \rightarrow \infty} \left[\int_{\mathcal{X}} u_{n_k}(x) dP(x) - \beta \log \left(\int_{\mathcal{Y}} e^{-\frac{v_{n_k}(y)}{\beta}} dQ(y) \right) \right. \\ &\quad \left. - \alpha \log \left(\int_{\mathcal{X} \times \mathcal{Y}} e^{\frac{u_{n_k}(x) + v_{n_k}(y) - c(x, y)}{\alpha}} dP \otimes Q(x, y) \right) \right] \end{aligned} \quad (95)$$

$$\begin{aligned} &\leq \lim_{k \rightarrow \infty} \left[\int_{\mathcal{X}} u_{n_k}(x) dP(x) \right] - \beta \liminf_{k \rightarrow \infty} \left[\log \left(\int_{\mathcal{Y}} e^{-\frac{v_{n_k}(y)}{\beta}} dQ(y) \right) \right] \\ &\quad - \alpha \liminf_{k \rightarrow \infty} \left[\log \left(\int_{\mathcal{X} \times \mathcal{Y}} e^{\frac{u_{n_k}(x) + v_{n_k}(y) - c(x, y)}{\alpha}} dP \otimes Q(x, y) \right) \right]. \end{aligned} \quad (96)$$

It follows from the weak* convergence of (u_{n_k}) to $\bar{u}$ that

$$\lim_{k \rightarrow \infty} \left[\int_{\mathcal{X}} u_{n_k}(x) dP(x) \right] = \int_{\mathcal{X}} \bar{u}(x) dP(x). \quad (97)$$

Combining (93), (94), (96) and the previous equality yields

$$\begin{aligned} \sup_{u \in L^\infty(\mathcal{X}), v \in L^\infty(\mathcal{Y})} D_{\alpha, \beta}(u, v) &\leq \int_{\mathcal{X}} \bar{u}(x) dP(x) - \beta \log \left(\int_{\mathcal{Y}} e^{-\frac{\bar{v}(y)}{\beta}} dQ(y) \right) \\ &\quad - \alpha \log \left(\int_{\mathcal{X} \times \mathcal{Y}} e^{\frac{\bar{u}(x) + \bar{v}(y) - c(x, y)}{\alpha}} dP \otimes Q(x, y) \right). \end{aligned} \quad (98)$$

Equivalently,

$$\sup_{u \in L^\infty(\mathcal{X}), v \in L^\infty(\mathcal{Y})} D_{\alpha, \beta}(u, v) \leq D_{\alpha, \beta}(\bar{u}, \bar{v}), \quad (99)$$

which follows from the definition of the supremum. Hence, $(\bar{u}, \bar{v})$ is a maximizer of $D_{\alpha, \beta}$, which concludes the proof that the supremum in (3) is attained.

It remains to show that any $(\hat{u}, \hat{v}) \in L^\infty(\mathcal{X}) \times L^\infty(\mathcal{Y})$ satisfying (10) and (11) is a maximizer of $D_{\alpha, \beta}$ in (3). This follows directly from Corollary B.1, since simultaneous satisfaction of (10) and (11) implies that $(\hat{u}, \hat{v})$ is a global maximizer of $(u, v) \mapsto D_{\alpha, \beta}(u, v)$. This completes the proof.

C Primal Attainment

C.1 Theorem C.1

Theorem C.1. *The infimum in problem (2) is attained at a unique measure $\hat{\pi} \in \Pi(P, \star)$. Moreover, $\hat{\pi}$ is absolutely continuous with respect to $P \otimes Q$, and its density satisfies*

$$\frac{d\hat{\pi}}{dP \otimes Q}(x, y) = e^{\frac{\hat{u}(x)}{\alpha}} e^{-\frac{c(x, y)}{\alpha}} e^{\frac{\hat{v}(y)}{\alpha}}, \quad (100)$$

where $(\hat{u}, \hat{v})$ is the unique pair of maximizing potentials satisfying (10) and (11).

The result is a consequence of Theorem 5.1: the entropic penalty $\alpha D_{\text{KL}}(\pi \| P \otimes Q)$ forces any coupling with finite primal value to be absolutely continuous with respect to $P \otimes Q$, and the first-order optimality conditions in the dual potentials then pin down the Gibbs-type density (100). Uniqueness follows from the strict convexity of the KL penalty. Let $(\hat{u}, \hat{v}) \in L^\infty(\mathcal{X}) \times L^\infty(\mathcal{Y})$ be a pair of maximizing potentials for problem (3) satisfying (10) and (11). Define $\hat{\pi}$ as in (100). Let $\hat{\rho}$ denote the density $\frac{d\hat{\pi}}{dP \otimes Q}$ and $\hat{s}$ denote the density $\frac{d\hat{\pi}_\#}{dQ}$. Condition (10) ensures that

$$\int_{\mathcal{Y}} \hat{\rho}(x, y) dQ(y) \stackrel{a.s.}{=} 1. \quad (101)$$

Thus, for any integrable function $f \in L^1(\mathcal{X})$, Fubini's theorem [Rudin, 1987, Theorem 8.8] yields

$$\int_{\mathcal{X} \times \mathcal{Y}} f(x) d\hat{\pi}(x, y) = \int_{\mathcal{X}} f(x) \left(\int_{\mathcal{Y}} \hat{\rho}(x, y) dQ(y) \right) dP(x) = \int_{\mathcal{X}} f(x) dP(x). \quad (102)$$

Hence, the measure $\hat{\pi}$ is a probability measure with first marginal $\hat{\pi}_\#^{\mathcal{X}} = P$, which implies $\hat{\pi} \in \Pi(P, \star)$. Besides, the optimality condition (11) implies

$$\int_{\mathcal{Y}} e^{\frac{-\hat{v}(y)}{\beta}} dQ(y) = 1, \text{ and } \hat{s}(y) \stackrel{a.s.}{=} e^{\frac{-\hat{v}(y)}{\beta}}. \quad (103)$$

From (101) and (103), the dual objective $D_{\alpha, \beta}(\hat{u}, \hat{v})$, with $D_{\alpha, \beta}$ in (3), can be computed as follows

$$\begin{aligned} D_{\alpha, \beta}(\hat{u}, \hat{v}) &= \int_{\mathcal{X}} \hat{u}(x) dP(x) - \beta \log \left(\int_{\mathcal{Y}} e^{\frac{-\hat{v}(y)}{\beta}} dQ(y) \right) \\ &\quad - \alpha \log \left(\int_{\mathcal{X} \times \mathcal{Y}} e^{\frac{\hat{u}(x) + \hat{v}(y) - c(x, y)}{\alpha}} dP \otimes Q(x, y) \right) \end{aligned} \quad (104)$$

$$= \int_{\mathcal{X}} \hat{u}(x) dP(x) - \beta \log(1) - \alpha \log(1) = \int_{\mathcal{X}} \hat{u}(x) dP(x). \quad (105)$$

From (2), the primal objective value at $\hat{\pi}$ is defined via

$$\begin{aligned} P_{\alpha,\beta}(\hat{\pi}) &= \int_{\mathcal{X} \times \mathcal{Y}} (c(x,y)\hat{\rho}(x,y) + \alpha(\hat{\rho}(x,y) \log(\hat{\rho}(x,y)) - \hat{\rho}(x,y) + 1)) dP \otimes Q(x,y) \\ &\quad + \beta \int_{\mathcal{Y}} (\hat{s}(y) \log(\hat{s}(y)) - \hat{s}(y) + 1) dQ(y). \end{aligned} \quad (106)$$

657 For all points $(x, y) \in \mathcal{X} \times \mathcal{Y}$, since $\hat{\rho}(x, y) = e^{\frac{\hat{u}(x) + \hat{v}(y) - c(x, y)}{\alpha}}$, the pointwise Fenchel-Young equality
658 [Zalinescu, 2002, Theorem 2.3.1] for the function $t \mapsto \alpha(t \log(t) - t + 1)$ states

$$\alpha(\hat{\rho}(x,y) \log(\hat{\rho}(x,y)) - \hat{\rho}(x,y) + 1) = \hat{\rho}(x,y)(\hat{u}(x) + \hat{v}(y) - c(x,y)) - \alpha(\hat{\rho}(x,y) - 1). \quad (107)$$

659 Substitution into (106) leads to

$$\begin{aligned} P_{\alpha,\beta}(\hat{\pi}) &= \int_{\mathcal{X} \times \mathcal{Y}} \hat{\rho}(x,y)(\hat{u}(x) + \hat{v}(y)) dP \otimes Q(x,y) - \int_{\mathcal{X} \times \mathcal{Y}} \alpha(\hat{\rho}(x,y) - 1) dP \otimes Q(x,y) \\ &\quad + \beta \int_{\mathcal{Y}} (\hat{s}(y) \log(\hat{s}(y)) - \hat{s}(y) + 1) dQ(y). \end{aligned} \quad (108)$$

660 As $\hat{\rho}$ is a probability density, the intergral term $\int_{\mathcal{X} \times \mathcal{Y}} (\hat{\rho}(x,y) - 1) dP \otimes Q(x,y)$ vanishes. Equation
661 (103) provides the relation $\log(\hat{s}) = -\hat{v}/\beta$. Pointwise Fenchel-Young equality [Zalinescu, 2002,
662 Theorem 2.3.1] for the second entropic term in (106) implies

$$\beta(\hat{s}(y) \log(\hat{s}(y)) - \hat{s}(y) + 1) = -\hat{s}(y)\hat{v}(y) - \beta(\hat{s}(y) - 1). \quad (109)$$

663 Linearity of the integral and Fubini's theorem [Rudin, 1987, Theorem 8.8] yield

$$\begin{aligned} P_{\alpha,\beta}(\hat{\pi}) &= \int_{\mathcal{X}} \hat{u}(x) \left(\int_{\mathcal{Y}} \hat{\rho}(x,y) dQ(y) \right) dP(x) + \int_{\mathcal{Y}} \hat{v}(y) \left(\int_{\mathcal{X}} \hat{\rho}(x,y) dP(x) \right) dQ(y) \\ &\quad - \int_{\mathcal{Y}} \hat{v}(y)\hat{s}(y) dQ(y) - \beta \int_{\mathcal{Y}} (\hat{s}(y) - 1) dQ(y). \end{aligned} \quad (110)$$

664 Definitions of the marginals $\hat{\pi}_{\#}^{\mathcal{X}}$ and $\hat{\pi}_{\#}^{\mathcal{Y}}$ coupled with identities (101) and (103) lead to

$$P_{\alpha,\beta}(\hat{\pi}) = \int_{\mathcal{X}} \hat{u}(x) dP(x) + \int_{\mathcal{Y}} \hat{v}(y)\hat{s}(y) dQ(y) - \int_{\mathcal{Y}} \hat{v}(y)\hat{s}(y) dQ(y) - \beta(1-1) \quad (111)$$

$$= \int_{\mathcal{X}} \hat{u}(x) dP(x) = D_{\alpha,\beta}(\hat{u}, \hat{v}), \quad (112)$$

665 where the last equality follows from (105). Therefore, from Lemma A.1, strong duality $P_{\alpha,\beta}(\hat{\pi}) =$
666 $D_{\alpha,\beta}(\hat{u}, \hat{v})$ holds and $\hat{\pi}$ is a minimizer for $P_{\alpha,\beta}$. Uniqueness of the minimizer $\hat{\pi}$ follows from the
667 strict convexity of the Kullback-Leibler divergence on its support.

668 C.2 Proof of Existence and Uniqueness of the Gauge-Fixed Pair in Theorem 5.1

669 The existence follows from Theorem 5.1. In particular, let $(\tilde{u}, \tilde{v})$ be a pair of maximizing potentials for
670 $D_{\alpha,\beta}$ in (3). Since $D_{\alpha,\beta}$ is Gateaux differentiable and concave, the first-order optimality conditions
671 in Lemma B.2 are satisfied. Define

$$u \triangleq \tilde{u} - \alpha \log(Z(\tilde{u}, \tilde{v})) - \beta \log(R(\tilde{v})), \quad (113)$$

$$v \triangleq \tilde{v} + \beta \log(R(\tilde{v})), \quad (114)$$

672 where

$$Z(f, g) \triangleq \int_{\mathcal{X} \times \mathcal{Y}} e^{\frac{f(x) + g(y) - c(x, y)}{\alpha}} dP \otimes Q(x, y); \text{ and } R(g) \triangleq \int_{\mathcal{Y}} e^{-\frac{g(y)}{\beta}} dQ(y). \quad (115)$$

673 From the translation invariance of $D_{\alpha, \beta}$ in Lemma B.1, the pair (u, v) defined in (113) and (114) is
 674 still a pair of maximizing potentials for $D_{\alpha, \beta}$. Moreover, (113), (114), and (115), imply

$$Z(u, v) = 1, \quad (116)$$

$$R(v) = 1. \quad (117)$$

675 From (116) and the first-order optimality condition in (41), it follows that

$$\int_{\mathcal{Y}} e^{\frac{u(x) + v(y) - c(x, y)}{\alpha}} dQ(y) \stackrel{a.s.}{=} 1. \quad (118)$$

676 Equation (118) is equivalent to (10). Similarly, from (116), and the first-order optimality condition in
 677 (42), it follows that

$$\int_{\mathcal{X}} e^{\frac{u(x) + v(y) - c(x, y)}{\alpha}} dP(x) \stackrel{a.s.}{=} e^{-\frac{v(y)}{\beta}}. \quad (119)$$

678 Equation (119), together with (116), is equivalent to (11). Hence, there exists at least one pair
 679 satisfying (10) and (11) simultaneously. This proves the existence.

680 It remains to prove the uniqueness of such a pair. Let (u_1, v_1) and (u_2, v_2) satisfy (10) and (11)
 681 simultaneously, that is

$$u_1 = \Phi(v_1), \quad v_1 = \Psi(u_1), \quad (120)$$

$$u_2 = \Phi(v_2), \quad v_2 = \Psi(u_2), \quad (121)$$

682 where Φ is in (4) and Ψ is in (5). Associate to each pair the coupling defined as follows

$$\pi_1(dx, dy) \triangleq \exp\left(\frac{u_1(x) + v_1(y) - c(x, y)}{\alpha}\right) P(dx) Q(dy), \quad (122)$$

$$\pi_2(dx, dy) \triangleq \exp\left(\frac{u_2(x) + v_2(y) - c(x, y)}{\alpha}\right) P(dx) Q(dy). \quad (123)$$

683 From Theorem C.1, (120) and (121), π_1 and π_2 are primal minimizers. Uniqueness of the primal
 684 minimizer (Theorem C.1) yields

$$\pi_1 = \pi_2, \quad (124)$$

685 which, together with (122) and (123), implies

$$\exp\left(\frac{u_1(x) + v_1(y) - c(x, y)}{\alpha}\right) \stackrel{a.s.}{=}_{P \otimes Q} \exp\left(\frac{u_2(x) + v_2(y) - c(x, y)}{\alpha}\right). \quad (125)$$

686 Taking logarithms in the previous equality gives

$$u_1(x) + v_1(y) \stackrel{a.s.}{\underset{P \otimes Q}{=}} u_2(x) + v_2(y). \quad (126)$$

687 From Fubini's theorem [Rudin, 1987, Theorem 8.8], there exists a set $\mathcal{Y}_0 \subset \mathcal{Y}$ such that $Q(\mathcal{Y}_0) = 1$
 688 and, for every $y \in \mathcal{Y}_0$, the equality in (126) holds for all $x \in \mathcal{X}$ almost surely with respect to P . Fix
 689 $y_0 \in \mathcal{Y}_0$. Then,

$$u_1(x) - u_2(x) = v_2(y_0) - v_1(y_0) \stackrel{a.s.}{\underset{P}{=}} K, \quad (127)$$

690 with

$$K \triangleq v_2(y_0) - v_1(y_0). \quad (128)$$

691 Combining (127) with (126) yields,

$$v_2(y) - v_1(y) \stackrel{a.s.}{\underset{Q}{=}} K. \quad (129)$$

692 Finally, combining (129) with the normalization in (75), which follows from (120) and (121), yields

$$1 = \int_{\mathcal{Y}} \exp\left(-\frac{v_2(y)}{\beta}\right) dQ(y) \quad (130)$$

$$= \int_{\mathcal{Y}} \exp\left(-\frac{v_1(y) + K}{\beta}\right) dQ(y) \quad (131)$$

$$= e^{-\frac{K}{\beta}} \int_{\mathcal{Y}} \exp\left(-\frac{v_1(y)}{\beta}\right) dQ(y) = e^{-\frac{K}{\beta}}. \quad (132)$$

693 Hence $K = 0$. Therefore, (127) and (129) imply

$$u_1 \stackrel{a.s.}{\underset{P}{=}} u_2, \quad v_1 \stackrel{a.s.}{\underset{Q}{=}} v_2. \quad (133)$$

694 This proves uniqueness of the pair satisfying (10) and (11). The proof is complete.

695 **D Approximation Error**

696 **D.1 Conditions for $SROT_Q = SROT$**

697 **Lemma D.1.** *Assume that Q has full support on $\mathcal{Y}$, and that for P -almost every $x \in \mathcal{X}$, the map*
 698 *$y \mapsto c(x, y)$ is continuous. Then*

$$SROT_Q = SROT. \quad (134)$$

699 *In particular, if $\mathcal{Y} = \{y_1, \dots, y_m\}$ is finite and $Q(\{y_j\}) > 0$ for every $j \in \{1, \dots, m\}$, then (134)*
 700 *holds.*

701 *Proof:* For all $x \in \mathcal{X}$, define

$$m(x) \triangleq \inf_{y \in \mathcal{Y}} c(x, y), \quad (135)$$

$$m_Q(x) \triangleq \operatorname{ess\,inf}_{y \sim Q} c(x, y). \quad (136)$$

702 The essential infimum is taken over a smaller class than the pointwise infimum, thus for all $x \in \mathcal{X}$, it
 703 holds that

$$m_Q(x) \geq m(x). \quad (137)$$

704 Consequently,

$$SROT_Q \geq SROT. \quad (138)$$

705 It remains to prove the reverse inequality. Fix $x \in \mathcal{X}$ be such that the map $y \mapsto c(x, y)$ is continuous.
 706 Given $\varepsilon > 0$, from the definition of $m(x)$, there exists $y_\varepsilon \in \mathcal{Y}$ such that

$$c(x, y_\varepsilon) < m(x) + \varepsilon. \quad (139)$$

707 The continuity of $y \mapsto c(x, y)$ at y_ε yields the existence of an open neighborhood U_ε of y_ε such that
 708 for all $y \in U_\varepsilon$, it follows that

$$c(x, y) < m(x) + \varepsilon. \quad (140)$$

709 Since Q has full support and U_ε is nonempty and open, it follows that

$$Q(U_\varepsilon) > 0. \quad (141)$$

710 From the definition of the essential infimum,

$$m_Q(x) \leq m(x) + \varepsilon. \quad (142)$$

711 Since $\varepsilon > 0$ is arbitrary, (142) yields

$$m_Q(x) = m(x) \quad (143)$$

712 for all x such that $y \mapsto c(x, y)$ is continuous. Therefore, (143) holds for $x \in \mathcal{X}$ almost surely with
 713 respect to P . Integrating with respect to P gives

$$\int_{\mathcal{X}} m_Q(x) dP(x) = \int_{\mathcal{X}} m(x) dP(x). \quad (144)$$

714 Since

$$SROT_Q = \int_{\mathcal{X}} m_Q(x) dP(x); \text{ and } SROT = \int_{\mathcal{X}} m(x) dP(x), \quad (145)$$

715 it follows that

$$SROT_Q = SROT. \quad (146)$$

716 Consider now the finite case. If $\mathcal{Y} = \{y_1, \dots, y_m\}$, then every map $y \mapsto c(x, y)$ is continuous.
 717 Moreover, the condition $Q(\{y_j\}) > 0$ for all $j \in \{1, \dots, m\}$ is equivalent to Q having full support
 718 on $\mathcal{Y}$. Hence the first part of the proof applies and yields

$$SROT_Q = SROT. \quad (147)$$

719 This completes the proof. ■

720 **D.2 Proof of Theorem 5.3**

721 The proof is divided into two parts, corresponding to the two assertions of the theorem.

722 **Proof of item 1:** Since $\alpha > 0$, every admissible coupling with finite value in (2) is absolutely
 723 continuous with respect to $P \otimes Q$. Therefore, for all such coupling $\pi \in \Pi(P, \star)$, it holds that

$$\int_{\mathcal{X} \times \mathcal{Y}} c(x, y) d\pi(x, y) \geq SROT_Q. \quad (148)$$

724 Since P and Q are probability measure then the Kullback–Leibler terms are non-negative, it then
 725 follows that

$$SROT_{\alpha, \beta} \geq SROT_Q. \quad (149)$$

726 It remains to prove the reverse inequality in the limit. For $\tau > 0$, define

$$Z_\tau(x) \triangleq \int_{\mathcal{Y}} e^{-\frac{c(x, z)}{\tau}} dQ(z), \quad (150)$$

$$K_x^\tau(dy) \triangleq \frac{e^{-\frac{c(x, y)}{\tau}}}{Z_\tau(x)} Q(dy), \quad (151)$$

$$\pi^\tau(dx, dy) \triangleq P(dx) K_x^\tau(dy). \quad (152)$$

727 Since c is bounded and Borel measurable, the map $x \mapsto Z_\tau(x)$ is measurable and strictly positive,
 728 and $x \mapsto K_x^\tau$ defines a probability kernel from $\mathcal{X}$ to $\mathcal{Y}$. Hence, the measure π^τ belongs to $\Pi(P, \star)$
 729 and satisfies $\pi^\tau \ll P \otimes Q$. Moreover,

$$\frac{d\pi^\tau}{dP \otimes Q}(x, y) = \frac{e^{-\frac{c(x, y)}{\tau}}}{Z_\tau(x)}. \quad (153)$$

Therefore,

$$\log \left(\frac{d\pi^\tau}{dP \otimes Q}(x, y) \right) = -\frac{c(x, y)}{\tau} - \log Z_\tau(x). \quad (154)$$

730 Since $|c(x, y)| \leq \|c\|_\infty$ for all $(x, y) \in \mathcal{X} \times \mathcal{Y}$, it holds that

$$e^{-\frac{\|c\|_\infty}{\tau}} \leq Z_\tau(x) \leq e^{\frac{\|c\|_\infty}{\tau}}. \quad (155)$$

731 Thus, from (154),

$$\left| \log \left(\frac{d\pi^\tau}{dP \otimes Q}(x, y) \right) \right| \leq \frac{2\|c\|_\infty}{\tau}. \quad (156)$$

732 Consequently,

$$D_{KL}(\pi^\tau \| P \otimes Q) \leq \frac{2\|c\|_\infty}{\tau}. \quad (157)$$

733 The data processing inequality yields

$$D_{KL}\left((\pi^\tau)_\# \|Q\right) \leq D_{KL}(\pi^\tau \| P \otimes Q) \leq \frac{2\|c\|_\infty}{\tau}. \quad (158)$$

734 Define

$$m_Q(x) \triangleq \operatorname{ess\,inf}_{y \sim Q} c(x, y). \quad (159)$$

735 Then

$$SROT_Q = \int_{\mathcal{X}} m_Q(x) dP(x). \quad (160)$$

736 Fix $\delta > 0$. For all $x \in \mathcal{X}$, define

$$A_{x,\delta} \triangleq \{y \in \mathcal{Y} : c(x, y) \leq m_Q(x) + \delta\}. \quad (161)$$

737 From the definition of the essential infimum, $Q(A_{x,\delta}) > 0$ for P -almost every x . Moreover,
738 $c(x, y) \geq m_Q(x)$ for Q -almost every y . Hence, for P -almost every x ,

$$e^{-\frac{m_Q(x)}{\tau}} \geq Z_\tau(x) \geq e^{-\frac{m_Q(x)+\delta}{\tau}} Q(A_{x,\delta}). \quad (162)$$

739 Taking logarithms and multiplying by $-\tau$ gives

$$m_Q(x) \leq -\tau \log Z_\tau(x) \leq m_Q(x) + \delta - \tau \log Q(A_{x,\delta}). \quad (163)$$

Letting $\tau \downarrow 0$ in (163), and then using the arbitrariness of $\delta > 0$, yields

$$-\tau \log Z_\tau(x) \xrightarrow{\tau \downarrow 0} m_Q(x) \quad (164)$$

740 for P -almost every x .

741 Moreover, from (153), it holds that

$$D_{KL}(K_x^\tau \| Q) = -\frac{1}{\tau} \int_{\mathcal{Y}} c(x, y) dK_x^\tau(y) - \log Z_\tau(x). \quad (165)$$

742 Therefore,

$$\int_{\mathcal{Y}} c(x, y) dK_x^\tau(y) = -\tau \log Z_\tau(x) - \tau D_{KL}(K_x^\tau \| Q) \leq -\tau \log Z_\tau(x). \quad (166)$$

743 On the other hand, since $c(x, y) \geq m_Q(x)$ for Q -almost every y , it holds that

$$\int_{\mathcal{Y}} c(x, y) dK_x^\tau(y) \geq m_Q(x). \quad (167)$$

744 Combining (164), (166), and (167), yields

$$\int_{\mathcal{Y}} c(x, y) dK_x^\tau(y) \xrightarrow{\tau \downarrow 0} m_Q(x) \quad (168)$$

745 for P -almost every x . Since

$$\left| \int_{\mathcal{Y}} c(x, y) dK_x^\tau(y) \right| \leq \|c\|_\infty, \quad (169)$$

the dominated convergence theorem yields

$$\int_{\mathcal{X} \times \mathcal{Y}} c(x, y) d\pi^\tau(x, y) = \int_{\mathcal{X}} \left(\int_{\mathcal{Y}} c(x, y) dK_x^\tau(y) \right) dP(x) \xrightarrow{\tau \downarrow 0} SROT_Q. \quad (170)$$

746 The optimality of the minimizer $\hat{\pi}_{\alpha, \beta}$ in (2) yields

$$SROT_{\alpha, \beta} \leq \int_{\mathcal{X} \times \mathcal{Y}} c(x, y) d\pi^\tau(x, y) + \alpha D_{KL}(\pi^\tau \| P \otimes Q) + \beta D_{KL}((\pi^\tau)_\#^\mathcal{Y} \| Q). \quad (171)$$

747 Using (157) and (158), it follows that

$$SROT_{\alpha, \beta} \leq \int_{\mathcal{X} \times \mathcal{Y}} c(x, y) d\pi^\tau(x, y) + \frac{2\|c\|_\infty(\alpha + \beta)}{\tau}. \quad (172)$$

748 Choosing $\tau = \sqrt{\alpha + \beta}$ in (172) and letting $(\alpha, \beta) \rightarrow (0, 0)$, (170) gives

$$\limsup_{\alpha, \beta \rightarrow 0} SROT_{\alpha, \beta} \leq SROT_Q, \quad (173)$$

749 which together with (149), proves (16). If $SROT_Q = SROT$, for instance under the assumptions of
750 Lemma D.1, then (16) gives $SROT_{\alpha, \beta} \rightarrow SROT$. This completes the proof of 1.

751 **Proof of item 2:** Assume first that (17) holds. The data processing inequality yields

$$D_{KL}(\tilde{\pi}_\#^\mathcal{Y} \| Q) \leq D_{KL}(\tilde{\pi} \| P \otimes Q) < +\infty. \quad (174)$$

752 The Kullback–Leibler terms in (2) are non-negative, it then follows that

$$SROT_{\alpha, \beta} \geq SROT. \quad (175)$$

753 Evaluating the objective defining $SROT_{\alpha, \beta}$ at $\tilde{\pi}$ gives

$$SROT_{\alpha, \beta} \leq \int_{\mathcal{X} \times \mathcal{Y}} c(x, y) d\tilde{\pi}(x, y) + \alpha D_{KL}(\tilde{\pi} \| P \otimes Q) + \beta D_{KL}(\tilde{\pi}_\#^\mathcal{Y} \| Q) \quad (176)$$

$$= SROT + \alpha D_{KL}(\tilde{\pi} \| P \otimes Q) + \beta D_{KL}(\tilde{\pi}_\#^\mathcal{Y} \| Q). \quad (177)$$

754 This proves (18), and concludes the proof.

755 D.3 Partial Limits in Value

756 **Proposition D.1.** *The following assertions hold.*

757 1. Fix $\beta > 0$. Assume that the problem defining $SROT_{0, \beta}$ admits a minimizer $\tilde{\pi}_\beta \in \Pi(P, \star)$
758 satisfying

$$D_{KL}(\tilde{\pi}_\beta \| P \otimes Q) < +\infty. \quad (178)$$

759 Then

$$0 \leq SROT_{\alpha, \beta} - SROT_{0, \beta} \leq \alpha D_{KL}(\tilde{\pi}_\beta \| P \otimes Q). \quad (179)$$

760 2. Fix $\alpha > 0$. Assume that the problem defining $SROT_{\alpha,0}$ admits a minimizer $\tilde{\pi}_\alpha \in \Pi(P, \star)$.
 761 Then $D_{KL}((\tilde{\pi}_\alpha)^\mathcal{Y}_\# \| Q) < +\infty$ and

$$0 \leq SROT_{\alpha,\beta} - SROT_{\alpha,0} \leq \beta D_{KL}((\tilde{\pi}_\alpha)^\mathcal{Y}_\# \| Q). \quad (180)$$

762 *Proof.* Consider first the bound (179). Fix $\beta > 0$ and assume that the problem defining $SROT_{0,\beta}$
 763 admits a minimizer $\tilde{\pi}_\beta \in \Pi(P, \star)$ satisfying (178). Since the additional term $\alpha D_{KL}(\pi \| P \otimes Q)$ is
 764 non-negative, it holds that

$$SROT_{\alpha,\beta} \geq SROT_{0,\beta}. \quad (181)$$

765 Evaluating the objective defining $SROT_{\alpha,\beta}$ at $\tilde{\pi}_\beta$ yields

$$SROT_{\alpha,\beta} \leq \int_{\mathcal{X} \times \mathcal{Y}} c(x, y) d\tilde{\pi}_\beta(x, y) + \alpha D_{KL}(\tilde{\pi}_\beta \| P \otimes Q) + \beta D_{KL}((\tilde{\pi}_\beta)^\mathcal{Y}_\# \| Q) \quad (182)$$

$$= SROT_{0,\beta} + \alpha D_{KL}(\tilde{\pi}_\beta \| P \otimes Q), \quad (183)$$

766 where the last equality follows from the optimality of $\tilde{\pi}_\beta$ for the problem defining $SROT_{0,\beta}$. This
 767 proves (179).

768 Consider now the bound (180). Fix $\alpha > 0$ and assume that the problem defining $SROT_{\alpha,0}$ admits a
 769 minimizer $\tilde{\pi}_\alpha \in \Pi(P, \star)$. Since $P \otimes Q \in \Pi(P, \star)$ and c is bounded,

$$SROT_{\alpha,0} \leq \int_{\mathcal{X} \times \mathcal{Y}} c(x, y) d(P \otimes Q)(x, y) < +\infty. \quad (184)$$

770 From the optimality of $\tilde{\pi}_\alpha$, it follows that

$$D_{KL}(\tilde{\pi}_\alpha \| P \otimes Q) < +\infty. \quad (185)$$

771 The data processing inequality then yields

$$D_{KL}((\tilde{\pi}_\alpha)^\mathcal{Y}_\# \| Q) \leq D_{KL}(\tilde{\pi}_\alpha \| P \otimes Q) < +\infty. \quad (186)$$

772 Since the additional term $\beta D_{KL}(\pi^\mathcal{Y}_\# \| Q)$ is non-negative, one has

$$SROT_{\alpha,\beta} \geq SROT_{\alpha,0}. \quad (187)$$

773 Evaluating the objective defining $SROT_{\alpha,\beta}$ at $\tilde{\pi}_\alpha$ gives

$$SROT_{\alpha,\beta} \leq \int_{\mathcal{X} \times \mathcal{Y}} c(x, y) d\tilde{\pi}_\alpha(x, y) + \alpha D_{KL}(\tilde{\pi}_\alpha \| P \otimes Q) + \beta D_{KL}((\tilde{\pi}_\alpha)^\mathcal{Y}_\# \| Q) \quad (188)$$

$$= SROT_{\alpha,0} + \beta D_{KL}((\tilde{\pi}_\alpha)^\mathcal{Y}_\# \| Q). \quad (189)$$

774 This proves (180), and concludes the proof. $\square$

775 **D.4 Proof of Theorem D.1**

776 **Theorem D.1.** *Let $(\alpha_n, \beta_n)_{n \geq 1}$ be a sequence in $(0, +\infty)^2$ with*

$$(\alpha_n, \beta_n) \xrightarrow{n \rightarrow \infty} (0, 0). \quad (190)$$

777 *Assume that*

$$SROT_{\alpha_n, \beta_n} \xrightarrow{n \rightarrow \infty} SROT, \quad (191)$$

778 *and that the family $(\hat{\pi}_{\alpha_n, \beta_n})_{n \geq 1}$ is tight. Then every weak limit point of $(\hat{\pi}_{\alpha_n, \beta_n})_{n \geq 1}$ is a minimizer*
 779 *of the unregularized semi-relaxed problem (1). If this minimizer is unique and denoted π^* , then*
 780 *$\hat{\pi}_{\alpha_n, \beta_n} \rightharpoonup \pi^*$.*

Proof: Let $(\hat{\pi}_{\alpha_{n_k}, \beta_{n_k}})_{k \geq 1}$ be a weakly convergent subsequence of $(\hat{\pi}_{\alpha_n, \beta_n})_{n \geq 1}$, and let

$$\hat{\pi}_{\alpha_{n_k}, \beta_{n_k}} \rightharpoonup \bar{\pi}. \quad (192)$$

Since the projection map

$$(x, y) \in \mathcal{X} \times \mathcal{Y} \longmapsto x \in \mathcal{X} \quad (193)$$

is continuous, the first marginal of $\bar{\pi}$ is preserved under weak convergence. Given that $(\hat{\pi}_{\alpha_{n_k}, \beta_{n_k}})_{\#}^{\mathcal{X}} = P$ for all k , it follows that

$$\bar{\pi}_{\#}^{\mathcal{X}} = P. \quad (194)$$

Consequently,

$$\bar{\pi} \in \Pi(P, \star). \quad (195)$$

Given that c is bounded and lower semi-continuous, the function $c + \|c\|_{\infty}$ is nonnegative and lower semi-continuous. The Portmanteau theorem [Billingsley, 1999, Theorem 2.1] therefore yields

$$\int_{\mathcal{X} \times \mathcal{Y}} (c(x, y) + \|c\|_{\infty}) d\bar{\pi}(x, y) \leq \liminf_{k \rightarrow \infty} \int_{\mathcal{X} \times \mathcal{Y}} (c(x, y) + \|c\|_{\infty}) d\hat{\pi}_{\alpha_{n_k}, \beta_{n_k}}(x, y). \quad (196)$$

Since both $\bar{\pi}$ and $\hat{\pi}_{\alpha_{n_k}, \beta_{n_k}}$ are probability measures, this is equivalent to

$$\int_{\mathcal{X} \times \mathcal{Y}} c(x, y) d\bar{\pi}(x, y) \leq \liminf_{k \rightarrow \infty} \int_{\mathcal{X} \times \mathcal{Y}} c(x, y) d\hat{\pi}_{\alpha_{n_k}, \beta_{n_k}}(x, y). \quad (197)$$

781 The Kullback–Leibler terms in (2) are nonnegative, hence, for every k ,

$$\int_{\mathcal{X} \times \mathcal{Y}} c(x, y) d\hat{\pi}_{\alpha_{n_k}, \beta_{n_k}}(x, y) \leq SROT_{\alpha_{n_k}, \beta_{n_k}}. \quad (198)$$

782 Combining (197) and (198), and using the assumption

$$SROT_{\alpha_n, \beta_n} \xrightarrow{n \rightarrow \infty} SROT, \quad (199)$$

783 gives

$$\int_{\mathcal{X} \times \mathcal{Y}} c(x, y) d\bar{\pi}(x, y) \leq SROT. \quad (200)$$

784 On the other hand, since $\bar{\pi} \in \Pi(P, \star)$, the definition of the unregularized semi-relaxed problem in (1)
 785 gives

$$\int_{\mathcal{X} \times \mathcal{Y}} c(x, y) d\bar{\pi}(x, y) \geq SROT. \quad (201)$$

786 Therefore,

$$\int_{\mathcal{X} \times \mathcal{Y}} c(x, y) d\bar{\pi}(x, y) = SROT, \quad (202)$$

787 which means that $\bar{\pi}$ is a minimizer of the unregularized semi-relaxed problem in (1). This proves that
 788 every weak limit point of $(\hat{\pi}_{\alpha_n, \beta_n})_{n \geq 1}$ is a minimizer of (1).

789 Assume now that this minimizer is unique, and denote it by π^* . By tightness and Prokhorov's theorem,
 790 the sequence $(\hat{\pi}_{\alpha_n, \beta_n})_{n \geq 1}$ is relatively compact for the weak topology. Hence every subsequence
 791 admits a weakly convergent subsequence. From the first part of the proof, every weak limit point is
 792 a minimizer of (1); by uniqueness, every weak limit point is equal to π^* . Consequently, the whole
 793 sequence converges weakly to π^* , that is,

$$\hat{\pi}_{\alpha_n, \beta_n} \rightharpoonup \pi^*. \quad (203)$$

794 This completes the proof. ■

795 **D.5 Proof of Theorem 5.4**

For all $i \in \{1, \dots, n\}$, recall that, the set Γ_i is defined by

$$\Gamma_i \triangleq \{j \in \{1, \dots, m\} : c(x_i, y_j) = m_i\}, \quad (204)$$

where

$$m_i \triangleq \min_{1 \leq j \leq m} c(x_i, y_j). \quad (205)$$

Since the first marginal of every $\pi \in \Pi(P, \star)$ is equal to P , it holds that

$$\sum_{j=1}^m \pi(x_i, y_j) = p_i, \quad i \in \{1, \dots, n\}. \quad (206)$$

Therefore, for all $\pi \in \Pi(P, \star)$,

$$\int_{\mathcal{X} \times \mathcal{Y}} c(x, y) d\pi(x, y) = \sum_{i=1}^n \sum_{j=1}^m c(x_i, y_j) \pi(x_i, y_j) \quad (207)$$

$$= \sum_{i=1}^n p_i m_i + \sum_{i=1}^n \sum_{j=1}^m (c(x_i, y_j) - m_i) \pi(x_i, y_j). \quad (208)$$

Since $c(x_i, y_j) - m_i \geq 0$ for all i, j , it follows that

$$SROT = \sum_{i=1}^n p_i m_i. \quad (209)$$

Moreover, equality is attained if and only if

$$\pi(x_i, y_j) = 0 \quad \text{for all } i \in \{1, \dots, n\} \text{ and } j \notin \Gamma_i. \quad (210)$$

796 Hence $\mathcal{O}$ is exactly the set of minimizers of (1).

Furthermore, for all $\pi \in \Pi(P, \star)$,

$$\int_{\mathcal{X} \times \mathcal{Y}} c(x, y) d\pi(x, y) - SROT = \sum_{i=1}^n \sum_{j=1}^m (c(x_i, y_j) - m_i) \pi(x_i, y_j). \quad (211)$$

For $j \in \Gamma_i$, the coefficient $c(x_i, y_j) - m_i$ vanishes. For $j \notin \Gamma_i$, the definition of Δ in (19) gives

$$c(x_i, y_j) - m_i \geq \Delta. \quad (212)$$

Consequently,

$$\int_{\mathcal{X} \times \mathcal{Y}} c(x, y) d\pi(x, y) - SROT \geq \Delta \sum_{i=1}^n \sum_{j \notin \Gamma_i} \pi(x_i, y_j). \quad (213)$$

Applying (213) to $\pi = \hat{\pi}_{\alpha, \beta}$, and using the non-negativity of the Kullback–Leibler terms in (2), yields

$$\int_{\mathcal{X} \times \mathcal{Y}} c(x, y) d\hat{\pi}_{\alpha, \beta}(x, y) \leq SROT_{\alpha, \beta}. \quad (214)$$

Hence,

$$\sum_{i=1}^n \sum_{j \notin \Gamma_i} \hat{\pi}_{\alpha, \beta}(x_i, y_j) \leq \frac{1}{\Delta} (SROT_{\alpha, \beta} - SROT). \quad (215)$$

For all $i \in \{1, \dots, n\}$, fix an index $j_i \in \Gamma_i$. Define $\tilde{\pi}$ by

$$\tilde{\pi}(x_i, y_j) = 0, \quad j \notin \Gamma_i, \quad (216)$$

$$\tilde{\pi}(x_i, y_j) = \hat{\pi}_{\alpha, \beta}(x_i, y_j), \quad j \in \Gamma_i \setminus \{j_i\}, \quad (217)$$

$$\tilde{\pi}(x_i, y_{j_i}) = \hat{\pi}_{\alpha, \beta}(x_i, y_{j_i}) + \sum_{j \notin \Gamma_i} \hat{\pi}_{\alpha, \beta}(x_i, y_j). \quad (218)$$

By construction, $\tilde{\pi} \in \Pi(P, \star)$ and

$$\tilde{\pi}(x_i, y_j) = 0, \quad i \in \{1, \dots, n\}, j \notin \Gamma_i. \quad (219)$$

Thus $\tilde{\pi} \in \mathcal{O}$. Moreover,

$$\|\hat{\pi}_{\alpha, \beta} - \tilde{\pi}\|_1 = 2 \sum_{i=1}^n \sum_{j \notin \Gamma_i} \hat{\pi}_{\alpha, \beta}(x_i, y_j). \quad (220)$$

Combining (215) and (220) gives

$$\inf_{\pi \in \mathcal{O}} \|\hat{\pi}_{\alpha, \beta} - \pi\|_1 \leq \frac{2}{\Delta} (SROT_{\alpha, \beta} - SROT). \quad (221)$$

Assume now that every row admits a unique minimizer. Then, for all $i \in \{1, \dots, n\}$, the set Γ_i is a singleton. Hence $\mathcal{O} = \{\pi^*\}$, where

$$\pi^*(x_i, y_j) = \begin{cases} p_i, & \text{if } j \in \Gamma_i, \\ 0, & \text{otherwise.} \end{cases} \quad (222)$$

Since $\mathcal{O} = \{\pi^*\}$, inequality (221) yields

$$\|\hat{\pi}_{\alpha, \beta} - \pi^*\|_1 \leq \frac{2}{\Delta} (SROT_{\alpha, \beta} - SROT). \quad (223)$$

It remains to prove the second inequality in the statement. Since $\mathcal{X}$ and $\mathcal{Y}$ are finite and $q_j > 0$ for all $j \in \{1, \dots, m\}$, the measure π^* is absolutely continuous with respect to $P \otimes Q$. Hence,

$$D_{KL}(\pi^* \| P \otimes Q) < +\infty. \quad (224)$$

Therefore, Assertion 2 of Theorem 5.3, applied with $\tilde{\pi} = \pi^*$, yields

$$SROT_{\alpha, \beta} - SROT \leq \alpha D_{KL}(\pi^* \| P \otimes Q) + \beta D_{KL}((\pi^*)_{\#}^{\mathcal{Y}} \| Q). \quad (225)$$

Combining (223) and (225) gives

$$\|\hat{\pi}_{\alpha, \beta} - \pi^*\|_1 \leq \frac{2}{\Delta} (SROT_{\alpha, \beta} - SROT) \quad (226)$$

$$\leq \frac{2}{\Delta} \left(\alpha D_{KL}(\pi^* \| P \otimes Q) + \beta D_{KL}((\pi^*)_{\#}^{\mathcal{Y}} \| Q) \right). \quad (227)$$

797 This completes the proof.

Remark (Tightness of the L^1 bound). The geometric factor $2/\Delta$ appearing in Theorem 5.4 cannot be improved, as the following minimal example demonstrates. Consider the discrete spaces $\mathcal{X} = \{x_1\}$ and $\mathcal{Y} = \{y_1, y_2\}$ with cost

$$c(x_1, y_1) = 0, \quad c(x_1, y_2) = \Delta, \quad (228)$$

where $\Delta > 0$. Since y_1 is the unique pointwise minimizer, the row-separation gap defined in (19) equals Δ , and the unregularized minimizer of (1) is the deterministic coupling

$$\pi^*(x_1, y_1) = 1, \quad \pi^*(x_1, y_2) = 0. \quad (229)$$

Any regularized algorithm assigns a small positive mass to the suboptimal target; let the regularized coupling be

$$\hat{\pi}(x_1, y_1) = 1 - \epsilon, \quad \hat{\pi}(x_1, y_2) = \epsilon, \quad (230)$$

for some $\epsilon \in (0, 1)$. The L^1 distance to the exact minimizer is

$$\|\hat{\pi} - \pi^*\|_1 = |(1 - \epsilon) - 1| + |\epsilon - 0| = 2\epsilon, \quad (231)$$

while the cost excess is

$$\int_{\mathcal{X} \times \mathcal{Y}} c(x, y) d\hat{\pi}(x, y) - SROT = 0 \cdot (1 - \epsilon) + \Delta \cdot \epsilon - 0 = \Delta\epsilon. \quad (232)$$

The ratio is therefore

$$\frac{\|\hat{\pi} - \pi^*\|_1}{\int_{\mathcal{X} \times \mathcal{Y}} c(x, y) d\hat{\pi}(x, y) - SROT} = \frac{2\epsilon}{\Delta\epsilon} = \frac{2}{\Delta}, \quad (233)$$

independently of ϵ . This proves that the geometric factor $2/\Delta$ governing the relation between the L^1 norm and the primal cost excess in Theorem 5.4 cannot be improved.

800 E Convergence Guarantees

801 E.1 Proof of Theorem 5.2

Proof: **a revoir en detail quand meme, j'ai beaucoup refait. Il reste peut etre des typos ou autres imprecisions** The proof is organized as follows. First, two positive linear operators are introduced and their projective diameters shown to be bounded. Theorem 2.9 in Lemmens and Nussbaum [2014] then yields strict contractions in Hilbert's metric. Next, the operators Ξ and Γ are shown to be Lipschitz, and the one step map $\Gamma \circ \Xi$ is shown to be a strict contraction. Finally, the normalization built into Γ identifies a complete slice of the positive cone, and the Banach fixed point theorem yields geometric convergence of the scaled iterates. The identification of the limit with the normalized maximizing potentials concludes the argument. Define the positive linear operators

$$K : L^\infty(\mathcal{Y}) \rightarrow L^\infty(\mathcal{X}), \quad (Kb)(x) \triangleq \int_{\mathcal{Y}} e^{-\frac{c(x,y)}{\alpha}} b(y) dQ(y), \quad (234)$$

$$K^\top : L^\infty(\mathcal{X}) \rightarrow L^\infty(\mathcal{Y}), \quad (K^\top a)(y) \triangleq \int_{\mathcal{X}} e^{-\frac{c(x,y)}{\alpha}} a(x) dP(x). \quad (235)$$

Since c is bounded, it holds for all $(x, y) \in \mathcal{X} \times \mathcal{Y}$ that

$$e^{-\frac{\|c\|_\infty}{\alpha}} \leq e^{-\frac{c(x,y)}{\alpha}} \leq e^{\frac{\|c\|_\infty}{\alpha}}. \quad (236)$$

Hence, for every $b \in L_+^\infty(\mathcal{Y}) \setminus \{0\}$ and every $x \in \mathcal{X}$,

$$e^{-\frac{\|c\|_\infty}{\alpha}} \int_{\mathcal{Y}} b(y) dQ(y) \leq (Kb)(x) \leq e^{\frac{\|c\|_\infty}{\alpha}} \int_{\mathcal{Y}} b(y) dQ(y). \quad (237)$$

Since $\int_{\mathcal{Y}} b(y) dQ(y) > 0$, this yields $Kb \in L_{++}^\infty(\mathcal{X})$. The same argument gives $K^\top a \in L_{++}^\infty(\mathcal{Y})$ for every $a \in L_+^\infty(\mathcal{X}) \setminus \{0\}$. If $b_1, b_2 \in L_+^\infty(\mathcal{Y}) \setminus \{0\}$, then for every $x \in \mathcal{X}$,

$$\frac{(Kb_1)(x)}{(Kb_2)(x)} \leq e^{\frac{2\|c\|_\infty}{\alpha}} \frac{\int_{\mathcal{Y}} b_1(y) dQ(y)}{\int_{\mathcal{Y}} b_2(y) dQ(y)}, \quad (238)$$

$$\frac{(Kb_1)(x)}{(Kb_2)(x)} \geq e^{-\frac{2\|c\|_\infty}{\alpha}} \frac{\int_{\mathcal{Y}} b_1(y) dQ(y)}{\int_{\mathcal{Y}} b_2(y) dQ(y)}. \quad (239)$$

Taking the essential supremum and the essential infimum over $x \in \mathcal{X}$ yields

$$d_H(K b_1, K b_2) \leq \frac{4\|c\|_\infty}{\alpha}. \quad (240)$$

Hence the projective diameter of K is finite, with

$$\Delta(K) \leq \frac{4\|c\|_\infty}{\alpha}. \quad (241)$$

The same argument yields

$$\Delta(K^\top) \leq \frac{4\|c\|_\infty}{\alpha}. \quad (242)$$

Since $\Delta(K) < +\infty$ and $\Delta(K^\top) < +\infty$, Theorem 2.9 in Lemmens and Nussbaum [2014] applies to the linear maps

$$K : L^\infty(\mathcal{Y}) \rightarrow L^\infty(\mathcal{X}) \quad \text{and} \quad K^\top : L^\infty(\mathcal{X}) \rightarrow L^\infty(\mathcal{Y}), \quad (243)$$

with source and target cones $L_+^\infty(\mathcal{Y})$ and $L_+^\infty(\mathcal{X})$. It follows that K is a strict contraction, for Hilbert's metric, from the part $L_{++}^\infty(\mathcal{Y})$ of $L_+^\infty(\mathcal{Y})$ into the part $L_{++}^\infty(\mathcal{X})$ of $L_+^\infty(\mathcal{X})$, and that $K^\top$ is a strict contraction, for Hilbert's metric, from $L_{++}^\infty(\mathcal{X})$ into $L_{++}^\infty(\mathcal{Y})$, with contraction factors $\tanh(\Delta(K)/4)$ and $\tanh(\Delta(K^\top)/4)$, respectively. Hence, for all $b_1, b_2 \in L_{++}^\infty(\mathcal{Y})$ and all $a_1, a_2 \in L_{++}^\infty(\mathcal{X})$, it holds that

$$d_H(Kb_1, Kb_2) \leq \tanh\left(\frac{\Delta(K)}{4}\right) d_H(b_1, b_2) \leq \tanh\left(\frac{\|c\|_\infty}{\alpha}\right) d_H(b_1, b_2), \quad (244)$$

$$d_H(K^\top a_1, K^\top a_2) \leq \tanh\left(\frac{\Delta(K^\top)}{4}\right) d_H(a_1, a_2) \leq \tanh\left(\frac{\|c\|_\infty}{\alpha}\right) d_H(a_1, a_2). \quad (245)$$

The first scaled update satisfies

$$\Xi(b)(x) = \frac{1}{(Kb)(x)}, \quad (246)$$

for every $b \in L_{++}^\infty(\mathcal{Y})$ and every $x \in \mathcal{X}$, where Ξ is in (7). Moreover, the definition of Hilbert's projective metric [Lemmens and Nussbaum, 2014, Equation (2.2)] implies that, for every pair f, g in either $L_{++}^\infty(\mathcal{X})$ or $L_{++}^\infty(\mathcal{Y})$,

$$d_H(f^{-1}, g^{-1}) = d_H(f, g). \quad (247)$$

Consequently, for all $b_1, b_2 \in L_{++}^\infty(\mathcal{Y})$,

$$d_H(\Xi(b_1), \Xi(b_2)) = d_H((Kb_1)^{-1}, (Kb_2)^{-1}) \quad (248)$$

$$= d_H(Kb_1, Kb_2) \quad (249)$$

$$\leq \tanh\left(\frac{\|c\|_\infty}{\alpha}\right) d_H(b_1, b_2). \quad (250)$$

Let $a \in L_{++}^\infty(\mathcal{X})$. The second scaled update can be written as

$$\Gamma(a)(y) = \left(K^\top a(y)\right)^{-\frac{\beta}{\alpha+\beta}} \left(\int_{\mathcal{Y}} \left(K^\top a(y')\right)^{\frac{\alpha}{\alpha+\beta}} dQ(y')\right)^{\frac{\beta}{\alpha}}, \quad (251)$$

and $\Gamma(a) \in L_{++}^\infty(\mathcal{Y})$, where Γ is in (8). The second factor is a positive scalar independent of y . Lemma 2.1(iv) in Lemmens and Nussbaum [2014] therefore gives, for all $a_1, a_2 \in L_{++}^\infty(\mathcal{X})$,

$$d_H(\Gamma(a_1), \Gamma(a_2)) = d_H\left(\left(K^\top a_1\right)^{-\frac{\beta}{\alpha+\beta}}, \left(K^\top a_2\right)^{-\frac{\beta}{\alpha+\beta}}\right). \quad (252)$$

The map $f \mapsto f^{\frac{\beta}{\alpha+\beta}}$ is order-preserving and homogeneous of degree $\frac{\beta}{\alpha+\beta}$. Proposition 2.6 in Lemmens and Nussbaum [2014] then implies

$$d_H\left(f^{\frac{\beta}{\alpha+\beta}}, g^{\frac{\beta}{\alpha+\beta}}\right) \leq \frac{\beta}{\alpha+\beta} d_H(f, g) \quad (253)$$

for all $f, g \in L_{++}^\infty(\mathcal{Y})$. Together with (247), this yields

$$d_H(\Gamma(a_1), \Gamma(a_2)) \leq \frac{\beta}{\alpha + \beta} d_H(K^\top a_1, K^\top a_2) \quad (254)$$

$$\leq \frac{\beta}{\alpha + \beta} \tanh\left(\frac{\|c\|_\infty}{\alpha}\right) d_H(a_1, a_2). \quad (255)$$

Combining (250) and (255) shows that the one step map $T \triangleq \Gamma \circ \Xi$ satisfies, for all $b_1, b_2 \in L_{++}^\infty(\mathcal{Y})$,

$$d_H(T(b_1), T(b_2)) \leq \frac{\beta}{\alpha + \beta} \tanh\left(\frac{\|c\|_\infty}{\alpha}\right)^2 d_H(b_1, b_2). \quad (256)$$

It remains to identify a complete metric space left invariant under T . For every $a \in L_{++}^\infty(\mathcal{X})$ and every $y \in \mathcal{Y}$,

$$\Gamma(a)(y)^{-\alpha/\beta} = \frac{\left(K^\top a(y)\right)^{\frac{\alpha}{\alpha+\beta}}}{\int_{\mathcal{Y}} \left(K^\top a(y')\right)^{\frac{\alpha}{\alpha+\beta}} dQ(y')}. \quad (257)$$

Integration with respect to Q yields

$$\int_{\mathcal{Y}} \Gamma(a)(y)^{-\alpha/\beta} dQ(y) = 1. \quad (258)$$

Hence every iterate $b^{(l)}$ belongs to

$$\Sigma \triangleq \left\{ b \in L_{++}^\infty(\mathcal{Y}) : \int_{\mathcal{Y}} b(y)^{-\alpha/\beta} dQ(y) = 1 \right\}. \quad (259)$$

Consider

$$q(b) \triangleq \left(\int_{\mathcal{Y}} b(y)^{-\alpha/\beta} dQ(y) \right)^{-\beta/\alpha}. \quad (260)$$

This map is homogeneous of degree 1. It is also continuous on $L_{++}^\infty(\mathcal{Y})$ for the L^∞ norm. Indeed, let b be in $L_{++}^\infty(\mathcal{Y})$ and $(b_n)_{n \geq 1}$ be a sequence in $L_{++}^\infty(\mathcal{Y})$ such that $b_n \rightarrow b$ in $L^\infty(\mathcal{Y})$, and set

$$m_b \triangleq \operatorname{ess\,inf}_Q b > 0. \quad (261)$$

There exists $n_0 \in \mathbb{N}$ such that $b_n(y) \geq m_b/2$ for Q -almost every $y \in \mathcal{Y}$ and every $n \geq n_0$. The map $t \mapsto t^{-\alpha/\beta}$ is Lipschitz on $[m_b/2, +\infty)$, hence

$$\left\| b_n^{-\alpha/\beta} - b^{-\alpha/\beta} \right\|_\infty \xrightarrow{n \rightarrow +\infty} 0. \quad (262)$$

Therefore

$$\int_{\mathcal{Y}} b_n(y)^{-\alpha/\beta} dQ(y) \xrightarrow{n \rightarrow +\infty} \int_{\mathcal{Y}} b(y)^{-\alpha/\beta} dQ(y), \quad (263)$$

and then $q(b_n) \rightarrow q(b)$. The cone $L_+^\infty(\mathcal{Y})$ is closed and normal in the Banach space $L^\infty(\mathcal{Y})$, and $L_{++}^\infty(\mathcal{Y})$ is the part containing the constant function 1. Lemma 2.1 in Nussbaum [2007], applied with the constant function 1, yields that (Σ, d_H) is a complete metric space. Since $b^{(0)} \equiv 1$ belongs to Σ and $T(\Sigma) \subset \Sigma$ follows from (258), the Banach fixed point theorem implies the existence of a unique fixed point $\hat{b} \in \Sigma$ and gives, for all $l \geq 0$,

$$d_H(b^{(l)}, \hat{b}) \leq \left(\frac{\beta}{\alpha + \beta} \tanh\left(\frac{\|c\|_\infty}{\alpha}\right)^2 \right)^l d_H(b^{(0)}, \hat{b}), \quad (264)$$

which is exactly (12). Define

$$\hat{a} \triangleq \Xi(\hat{b}). \quad (265)$$

Then, for every $l \geq 1$,

$$d_H(a^{(l)}, \hat{a}) = d_H(\Xi(b^{(l-1)}), \Xi(\hat{b})) \quad (266)$$

$$\leq \tanh\left(\frac{\|c\|_\infty}{\alpha}\right) d_H(b^{(l-1)}, \hat{b}) \quad (267)$$

$$\leq \tanh\left(\frac{\|c\|_\infty}{\alpha}\right) \rho_{(c, \alpha, \beta)}^{l-1} d_H(b^{(0)}, \hat{b}), \quad (268)$$

802 which is exactly (13).

Finally, define

$$\hat{u} \triangleq \alpha \log(\hat{a}) \in L^\infty(\mathcal{X}), \quad \hat{v} \triangleq \alpha \log(\hat{b}) \in L^\infty(\mathcal{Y}). \quad (269)$$

The fixed point relation $T(\hat{b}) = \hat{b}$ and the definition $\hat{a} = \Xi(\hat{b})$ are equivalent to

$$\hat{u} = \Phi(\hat{v}), \quad \hat{v} = \Psi(\hat{u}), \quad (270)$$

803 where Φ and Ψ are defined in (4) and (5). Corollary B.1 and Theorem 5.1 then imply that $\hat{u}$ and $\hat{v}$
 804 the unique maximizers for problem (3) that satisfy (10) and (11) simultaneously. Therefore, $\hat{a}$ and $\hat{b}$
 805 are the associated unique scaled maximizing potentials. ■

806 E.2 Corollary E.1

Corollary E.1. *The unscaled iterates $(u^{(l)}, v^{(l)})$ defined by (6) converge uniformly to $(\hat{u}, \hat{v})$, the unique pair of maximizers for problem (3) that satisfy (10) and (11) simultaneously, at a linear rate: for all $l \geq 0$,*

$$\|v^{(l)} - \hat{v}\|_\infty \leq \alpha (\rho_{(c, \alpha, \beta)})^l d_H(b^{(0)}, \hat{b}), \quad (271)$$

$$\|u^{(l+1)} - \hat{u}\|_\infty \leq \alpha (\rho_{(c, \alpha, \beta)})^l d_H(b^{(0)}, \hat{b}). \quad (272)$$

Proof: The proof is split into three parts; the first one deals with (271), the second one with (272), and the third one with (299). Let $(u^{(l)}, v^{(l)})$ be the unscaled iterates defined by (6), $(a^{(l)}, b^{(l)})$ be the scaled iterates defined by Algorithm 1, and $(\hat{u}, \hat{v})$ be the unique pair of maximizers satisfying (10) and (11) simultaneously. Recall that the two parametrizations are linked through

$$u^{(l)} = \alpha \log(a^{(l)}), \quad v^{(l)} = \alpha \log(b^{(l)}). \quad (273)$$

Hence, it holds that

$$\|v^{(l)} - \hat{v}\|_\infty = \alpha \left\| \log\left(\frac{b^{(l)}}{\hat{b}}\right) \right\|_\infty. \quad (274)$$

Define the pointwise ratio

$$r^{(l)}(y) \triangleq \frac{b^{(l)}(y)}{\hat{b}(y)}, \quad y \in \mathcal{Y}, \quad (275)$$

and its bounds

$$m^{(l)} \triangleq \inf_{y \in \mathcal{Y}} r^{(l)}(y), \quad M^{(l)} \triangleq \sup_{y \in \mathcal{Y}} r^{(l)}(y). \quad (276)$$

By definition of Hilbert's projective metric on $L_{++}^\infty(\mathcal{Y})$ [Lemmens and Nussbaum, 2014, Equation (2.2)], it holds that

$$d_H(b^{(l)}, \hat{b}) = \log\left(\frac{M^{(l)}}{m^{(l)}}\right). \quad (277)$$

It remains to show that

$$m^{(l)} \leq 1 \leq M^{(l)}. \quad (278)$$

Since the iterates $b^{(l)}$ are in Σ for all l and $\hat{b} \in \Sigma$, where Σ is in (259), the same gauge constraint as in (258) holds

$$\int_{\mathcal{Y}} (b^{(l)}(y))^{-\alpha/\beta} dQ(y) = 1, \quad \text{and} \quad \int_{\mathcal{Y}} (\hat{b}(y))^{-\alpha/\beta} dQ(y) = 1. \quad (279)$$

The map $t \mapsto t^{-\alpha/\beta}$ is strictly decreasing on $(0, +\infty)$. Assume by contradiction that $M^{(l)} < 1$. Then $r^{(l)}(y) < 1$ for Q -a.e. y , i.e., $b^{(l)}(y) < \hat{b}(y)$ for Q -a.e. y , and the strict monotonicity of the integral implies

$$\int_{\mathcal{Y}} (b^{(l)}(y))^{-\alpha/\beta} dQ(y) > \int_{\mathcal{Y}} (\hat{b}(y))^{-\alpha/\beta} dQ(y), \quad (280)$$

which contradicts (279). Therefore $M^{(l)} \geq 1$. The case $m^{(l)} > 1$ is ruled out similarly. Thus, the inequalities in (278) are established. Combining (278) with $\|\log r^{(l)}\|_{\infty} = \max\{-\log m^{(l)}, \log M^{(l)}\}$ gives

$$\left\| \log \left(\frac{b^{(l)}}{\hat{b}} \right) \right\|_{\infty} = \max\{-\log m^{(l)}, \log M^{(l)}\} \quad (281)$$

$$\leq \log \left(\frac{M^{(l)}}{m^{(l)}} \right) \quad (282)$$

$$= d_{\text{H}}(b^{(l)}, \hat{b}), \quad (283)$$

where the last equality follows from (277). Plugging (283) into (274) yields

$$\|v^{(l)} - \hat{v}\|_{\infty} \leq \alpha d_{\text{H}}(b^{(l)}, \hat{b}). \quad (284)$$

Combining the previous inequality with (12) concludes the proof of (271). Regarding the proof of (272), recall from (6) that

$$u^{(l+1)} = \Phi(v^{(l)}), \quad \text{and} \quad \hat{u} = \Phi(\hat{v}), \quad (285)$$

where Φ is in (4). Let v, w be in $L^{\infty}(\mathcal{Y})$ and set $\delta \triangleq \|v - w\|_{\infty}$. Then, for Q -a.e. y in $\mathcal{Y}$, it holds that

$$-\delta \leq v(y) - w(y) \leq \delta \implies e^{-\delta/\alpha} \leq \exp\left(\frac{v(y) - w(y)}{\alpha}\right) \leq e^{\delta/\alpha}. \quad (286)$$

Multiplying by $\exp\left(\frac{w(y) - c(x, y)}{\alpha}\right)$ and integrating with respect to Q yields, for every $x \in \mathcal{X}$,

$$e^{-\delta/\alpha} \int_{\mathcal{Y}} \exp\left(\frac{w(y) - c(x, y)}{\alpha}\right) dQ(y) \leq \int_{\mathcal{Y}} \exp\left(\frac{v(y) - c(x, y)}{\alpha}\right) dQ(y) \quad (287)$$

$$\leq e^{\delta/\alpha} \int_{\mathcal{Y}} \exp\left(\frac{w(y) - c(x, y)}{\alpha}\right) dQ(y). \quad (288)$$

Applying $-\alpha$ times the logarithm to the previous inequalities yields, for every $x \in \mathcal{X}$,

$$\Phi(w)(x) - \delta \leq \Phi(v)(x) \leq \Phi(w)(x) + \delta, \quad (289)$$

hence

$$\|\Phi(v) - \Phi(w)\|_{\infty} \leq \|v - w\|_{\infty}. \quad (290)$$

Combining (285) with (290) yields

$$\|u^{(l+1)} - \hat{u}\|_{\infty} = \|\Phi(v^{(l)}) - \Phi(\hat{v})\|_{\infty} \leq \|v^{(l)} - \hat{v}\|_{\infty}. \quad (291)$$

Together with (271), this implies

$$\|u^{(l+1)} - \hat{u}\|_{\infty} \leq \alpha d_{\text{H}}(b^{(l)}, \hat{b}). \quad (292)$$

807 Finally, applying (12) concludes the proof of (272). ■

808 **E.3 Corollary E.2**

Corollary E.2. Let $(u^{(l)}, v^{(l)})_{l \geq 0}$ be the unscaled iterates defined by (6). The sequence $(D_{\alpha, \beta}(u^{(l)}, v^{(l)}))_{l \geq 0}$ is non-decreasing and converges to $P_{\alpha, \beta}(\hat{\pi})$ at a linear rate: for all $l \geq 0$,

$$0 \leq P_{\alpha, \beta}(\hat{\pi}) - D_{\alpha, \beta}(u^{(l)}, v^{(l)}) \leq 4\alpha (\rho_{(\mathbf{c}, \alpha, \beta)})^l d_H(b^{(0)}, \hat{b}), \quad (293)$$

809 where $\rho_{(\mathbf{c}, \alpha, \beta)}$ is defined in (14) and $\hat{b} \triangleq e^{\hat{v}/\alpha}$.

Proof: From Corollary B.1, each alternating maximization step does not decrease the value of $D_{\alpha, \beta}$, hence $D_{\alpha, \beta}(u^{(l)}, v^{(l)}) \leq D_{\alpha, \beta}(u^{(l+1)}, v^{(l+1)})$ for all $l \geq 0$. Strong duality, established in Theorem C.1, provides the left-hand inequality in (293). The right-hand inequality rests on the following observation: for any measurable bounded functions f and g on a probability space (Ω, μ) , it holds that

$$\left| \log \int_{\Omega} e^{f(t)} d\mu(t) - \log \int_{\Omega} e^{g(t)} d\mu(t) \right| \leq \|f - g\|_{\infty}. \quad (294)$$

Indeed, $\int e^{f(t)} d\mu(t) \leq e^{\|f - g\|_{\infty}} \int e^{g(t)} d\mu(t)$, from which it can be deduced that

$$\log \int_{\Omega} e^{f(t)} d\mu(t) - \log \int_{\Omega} e^{g(t)} d\mu(t) \leq \|f - g\|_{\infty}. \quad (295)$$

The reverse inequality is obtained symmetrically. From definition (3), the gap decomposes as

$$\begin{aligned} D_{\alpha, \beta}(\hat{u}, \hat{v}) - D_{\alpha, \beta}(u^{(l)}, v^{(l)}) &= \int_{\mathcal{X}} (\hat{u} - u^{(l)}) dP \\ &\quad - \beta \left[\log \int_{\mathcal{Y}} e^{-\hat{v}(y)/\beta} dQ(y) - \log \int_{\mathcal{Y}} e^{-v^{(l)}(y)/\beta} dQ(y) \right] \\ &\quad - \alpha \left[\log \int_{\mathcal{X} \times \mathcal{Y}} e^{(\hat{u}(x) + \hat{v}(y) - \mathbf{c}(x, y))/\alpha} dP \otimes Q(x, y) \right. \\ &\quad \left. - \log \int_{\mathcal{X} \times \mathcal{Y}} e^{(u^{(l)}(x) + v^{(l)}(y) - \mathbf{c}(x, y))/\alpha} dP \otimes Q(x, y) \right]. \end{aligned} \quad (296)$$

Applying (294) to the second and third terms of (296), and using the fact that P is a probability measure for the first term, yields

$$D_{\alpha, \beta}(\hat{u}, \hat{v}) - D_{\alpha, \beta}(u^{(l)}, v^{(l)}) \leq 2 \|\hat{u} - u^{(l)}\|_{\infty} + 2 \|\hat{v} - v^{(l)}\|_{\infty}. \quad (297)$$

810 Applying (272) and (271) to the right-hand side of (297) concludes the proof. ■

811 **E.4 Corollary E.3**

Corollary E.3. Let $(u^{(l)}, v^{(l)})_{l \geq 0}$ be the unscaled iterates defined by (6), and let $M \triangleq \max(2, 3 \frac{\beta+1}{\alpha+\beta})$. For each $l \geq 0$, define the measure $\pi^{(l)}$ on $\mathcal{X} \times \mathcal{Y}$ through

$$\frac{d\pi^{(l)}}{dP \otimes Q}(x, y) \triangleq \exp\left(\frac{u^{(l+1)}(x) + v^{(l)}(y) - \mathbf{c}(x, y)}{\alpha}\right). \quad (298)$$

Then $\pi^{(l)} \in \Pi(P, \star)$ for all $l \geq 0$, and the sequence $(d\pi^{(l)}/dP \otimes Q)$ converges to $d\hat{\pi}/dP \otimes Q$ in $L^{\infty}(\mathcal{X} \times \mathcal{Y})$ at a linear speed, that is

$$\left\| \frac{d\pi^{(l)}}{dP \otimes Q} - \frac{d\hat{\pi}}{dP \otimes Q} \right\|_{\infty} \leq 2e^{(2M+2)\|\mathbf{c}\|_{\infty}/\alpha} (\rho_{(\mathbf{c}, \alpha, \beta)})^l d_H(b^{(0)}, \hat{b}), \quad (299)$$

812 where $\rho_{(\mathbf{c}, \alpha, \beta)}$ is defined in (14). In particular, $\pi^{(l)}$ converges to $\hat{\pi}$ in total variation at the same
813 linear rate.

Proof: The proof establishes successively that $\pi^{(l)}$ is in $\Pi(P, \star)$, that the relevant exponents are uniformly bounded, and that this bound combined with the linear convergence of the potentials yields (299). Since $u^{(l+1)} = \Phi(v^{(l)})$ by (6a), the definition (4) of Φ gives, for every $x \in \mathcal{X}$,

$$\begin{aligned} \int_{\mathcal{Y}} \frac{d\pi^{(l)}}{dP \otimes Q}(x, y) dQ(y) &= e^{u^{(l+1)}(x)/\alpha} \int_{\mathcal{Y}} e^{(v^{(l)}(y) - \mathbf{c}(x, y))/\alpha} dQ(y) \\ &= e^{u^{(l+1)}(x)/\alpha} \cdot e^{-u^{(l+1)}(x)/\alpha} = 1. \end{aligned} \quad (300)$$

Fubini's theorem [Rudin, 1987, Theorem 8.8] then implies that $\pi^{(l)}$ is a probability measure with first marginal $(\pi^{(l)})_{\#}^{\mathcal{X}} = P$, hence $\pi^{(l)} \in \Pi(P, \star)$. From (69), it holds that $\|v^{(l)}\|_{\infty} \leq M\|c\|_{\infty}$ for all $l \geq 1$, and from (68), $\|u^{(l+1)}\|_{\infty} \leq \|v^{(l)}\|_{\infty} + \|c\|_{\infty} \leq (M+1)\|c\|_{\infty}$ for all $l \geq 1$. The same estimates hold for $\hat{u}$ and $\hat{v}$, since $(\hat{u}, \hat{v})$ satisfies (10) and (11). Hence, for all $(x, y) \in \mathcal{X} \times \mathcal{Y}$ and all $l \geq 0$, the two arguments

$$\frac{u^{(l+1)}(x) + v^{(l)}(y) - c(x, y)}{\alpha} \quad \text{and} \quad \frac{\hat{u}(x) + \hat{v}(y) - c(x, y)}{\alpha} \quad (301)$$

are both bounded in absolute value by $(2M+2)\|c\|_{\infty}/\alpha$. For real numbers a and b with $|a|, |b| \leq R$, the mean value theorem gives $|e^a - e^b| \leq e^R|a - b|$. Applying this pointwise with $R = (2M+2)\|c\|_{\infty}/\alpha$ and using (301), for all $(x, y) \in \mathcal{X} \times \mathcal{Y}$ it holds that

$$\begin{aligned} & \left| \frac{d\pi^{(l)}}{dP \otimes Q}(x, y) - \frac{d\hat{\pi}}{dP \otimes Q}(x, y) \right| \\ & \leq \frac{e^{(2M+2)\|c\|_{\infty}/\alpha}}{\alpha} \left(|u^{(l+1)}(x) - \hat{u}(x)| + |v^{(l)}(y) - \hat{v}(y)| \right). \end{aligned} \quad (302)$$

Applying (272) and (271) to the previous expression yields

$$\left\| \frac{d\pi^{(l)}}{dP \otimes Q} - \frac{d\hat{\pi}}{dP \otimes Q} \right\|_{\infty} \leq 2e^{(2M+2)\|c\|_{\infty}/\alpha} \rho_{(\mathbf{c}, \alpha, \beta)}^l d_H(b^{(0)}, \hat{b}), \quad (303)$$

which gives (299). The total variation bound then follows from

$$\|\pi^{(l)} - \hat{\pi}\|_{TV} \triangleq \frac{1}{2} \int_{\mathcal{X} \times \mathcal{Y}} \left| \frac{d\pi^{(l)}}{dP \otimes Q} - \frac{d\hat{\pi}}{dP \otimes Q} \right| dP \otimes Q \leq \frac{1}{2} \left\| \frac{d\pi^{(l)}}{dP \otimes Q} - \frac{d\hat{\pi}}{dP \otimes Q} \right\|_{\infty} \quad (304)$$

814 since $P \otimes Q$ is a probability measure. ■

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